

UPSC Prelims

Environment & Ecology

Class No. 15

Pollution (final class)

- **System of Air Quality and Weather Forecasting and Research (SAFAR)**
- It is a national initiative introduced by the **Ministry of Earth Sciences (MoES)** (**not MoEFCC**).
- It is an integral part of India's first Air Quality Early Warning System operational in Delhi.
- SAFAR will measure the air quality of a metropolitan city.
- It also monitors all weather parameters like temperature, rainfall, humidity, wind speed, and wind direction, UV radiation, and solar radiation.
- The network has **multiple air quality monitoring stations** in each city, equipped with real-time, continuous monitors for various pollutants)

- The system is indigenously developed by the **Indian Institute of Tropical Meteorology (IITM), Pune** and is operationalised by the India Meteorological Department (IMD).
- It has a giant colour LED display that gives out real-time air quality index on a 24x7 basis with **colour-coding** (along with 72 hours advance forecast).
- The ultimate objective of the project is to increase awareness among the public regarding the air quality in their city.

- Pollutants monitored under SAFAR System include:

1. PM_{2.5},
2. PM₁₀,
3. Ozone,
4. Carbon Monoxide (CO),
5. Nitrogen Oxides (NO_x),
6. Sulphur Dioxide (SO₂),
7. Mercury
8. Benzene
9. Toluene (methylbenzene — used in paint thinners & as octane booster in gasoline engines),
10. Xylene (dimethylbenzene — used as a solvent in printing, rubber, and leather industry), and

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Degrees of Air Pollution

SAFAR AQI Scale		World AQI Scale	
Good	0-50	Good	0-50
Satisfactory	51-100	Satisfactory	50-100
Moderate	100-200	Unhealthy**	100-150
Poor	201-300	Unhealthy	150-200
Very Poor	301-400	Very Unhealthy	200-300
Severe	401-500	Severe	Over 300

- **National Air Quality Index (AQI)**
- AQI was launched by **MoEFCC** in April 2015 under ‘Swachh Bharat’.
- It helps the citizens to judge the air quality within her vicinity.
- There are **six AQI** categories, namely Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe.
- The AQI will consider **eight** pollutants
- PM10 and PM2.5
- NO2, SO2,
- Carbon monoxide (CO), Ozone (O3)
- Ammonia (NH3) and Lead (Pb)

- **National Air Quality Resource Framework of India (NARFI)**
- NARFI is an **information mechanism** to help decision-makers in government, municipalities, start-ups and in private sectors to address air pollution issues in different climatic zones of India.
- Developed by: **National Institute of Advanced Studies (NIAS)**, Bengaluru with support from Office of Principal Scientific Adviser.
- Will provide an all-inclusive guide to collecting air quality data, studying its impact and implementing science based solutions.
- Will have **5 modules** including Emission Inventory, Air Shed, and Mitigation; Impacts on Human Health and Agriculture etc.

- **Norms for thermal power plants (TPPs)**
- Context: Ministry of Environment, Forest and Climate Change (MoEFCC) gave two year extension to thermal power plants (TPPs) on SO₂ norms.
- **About TPP Norms**
- In 2015, MoEFCC notified environmental norms for particulate matter (**PM**), sulphur dioxides (**SO₂**), **nitrogen oxides** and **mercury** and **water use** for coal based TPPs.
- Initially deadline for implementation- 2017 was extended to 2022.

Coal TPPs contribute to over half SO₂ concentration, 30% oxides of nitrogen, 20% PM in the ambient air..

- Deadline for SO₂ further extended for all three categories of TPPs.
- However, deadline remains **same** for meeting **PM and NO_x standards**.
- Compliance with norms requires retrofitting existing TPPs with auxiliaries to control emissions such as **Flue Gas Desulphurisation (FGD)**, Selective Catalytic Reduction etc.
- FGD: A set of technologies used to remove SO₂ from sources of emissions, e.g., exhaust flue gases of fossil-fuel power plants.
- **Thermal power plants** are the **biggest users** of water among all industries.

SO₂ Norms for Thermal Power Plants (TPPs)

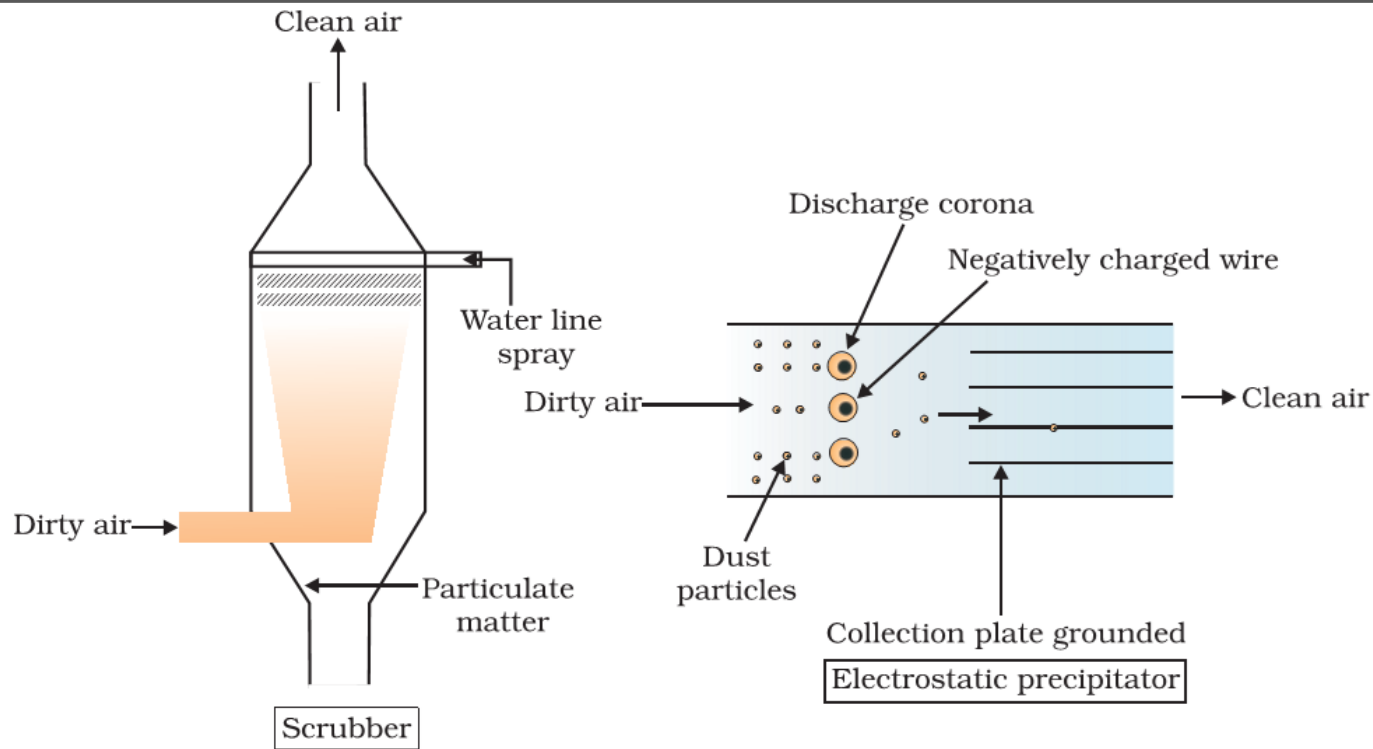
 Category	 Criteria	 Deadline for parameters other than SO ₂ emissions	 Deadline for SO ₂ emissions
A	Within 10 km radius of the National Capital Region (NCR) or cities having million-plus population .	End of 2022	End of 2024
B	Within 10 km radius of critically polluted areas or non-attainment cities (not meeting National Ambient Air Quality Standards).	End of 2023	End of 2025
C	Remaining plants	End of 2024	End of 2026



- **Electrostatic Precipitators (ESP)**
- Electrostatic precipitation can remove over **99 per cent of particulate matter** present in the exhaust.
- They are used in **boilers** and **furnaces**, thermal power plants, cement factories, steel plants, etc.
- In electrostatic precipitation, emanating dust is **charged with ions**, and the ionised particulate matter is collected at an oppositely charged surface.

- **Working**
- An electrostatic precipitator has **electrode wires** that are maintained at several thousand volts, which produce a corona that releases electrons.
- These electrons attach to dust particles giving them a **net negative charge**.
- The collecting plates are **grounded** (relatively positive charge) and attract the charged dust particles.
- The air velocity between the plates must be low enough to allow the dust to fall.
- The particles are removed from the collection surface by **occasional shaking or rapping**.

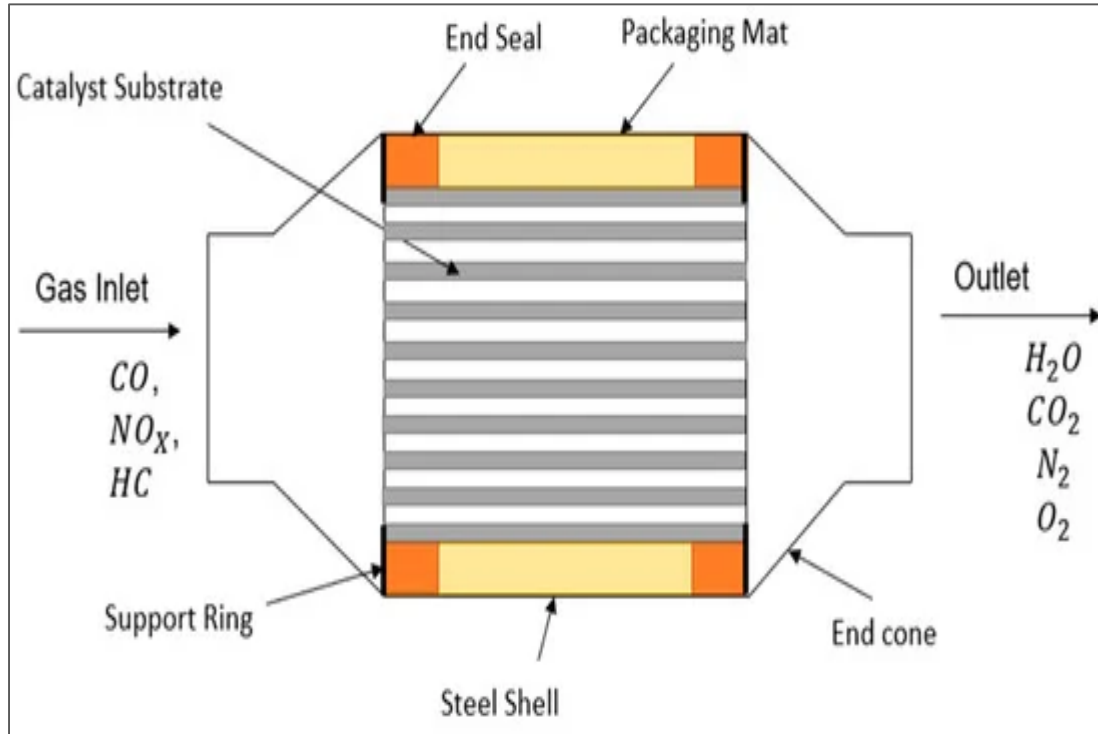
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- **Scrubbers**
- Scrubbers are **wet collectors**. They remove aerosols from a stream of gas either by collecting wet particles on a surface, or the particles are wetted by a scrubbing liquid.
- The particles get **trapped** as they travel from the supporting gaseous medium across the interface to the liquid scrubbing medium (this is similar to mucus in trachea trapping dust).
- A scrubber can remove gases like **sulphur dioxide**.
- **Inertial Collectors**
- Inertia of **Suspended Particulate Matter** in gas is higher than its solvent. As inertia is a function of the mass of the particulate matter, the device collects heavier particles more efficiently (centrifugation is the technique).
- ‘**Cyclone**’ is a common inertial collector used in gas cleaning plants.

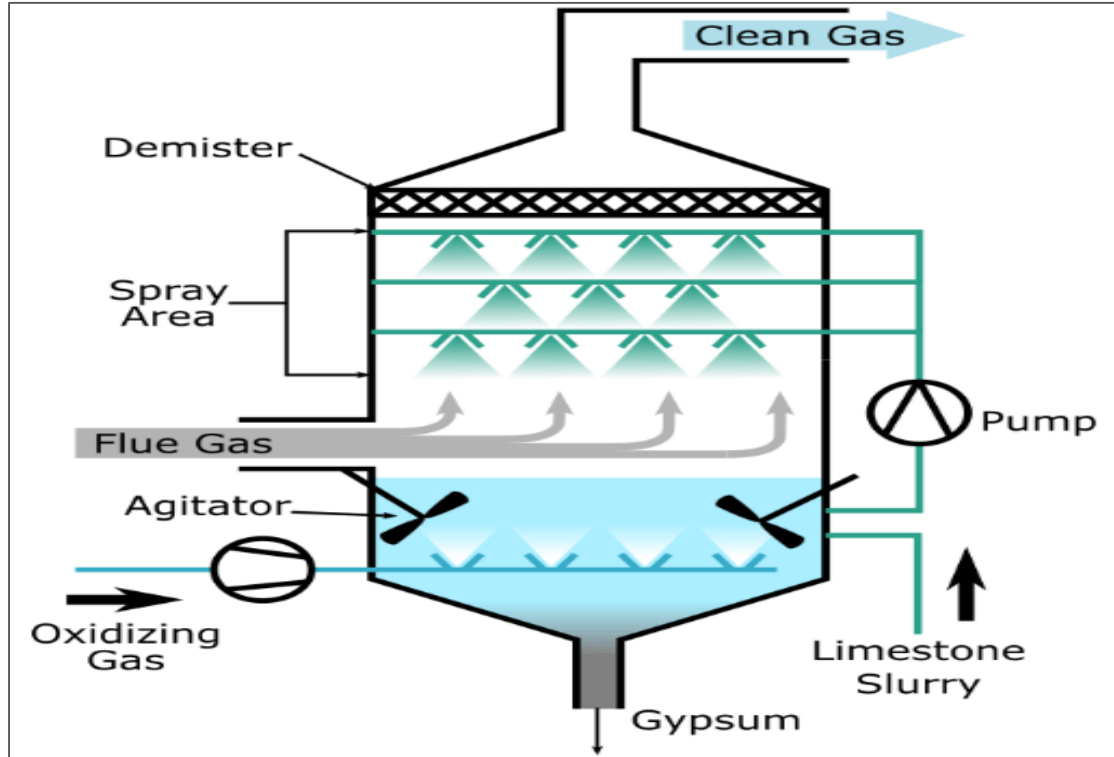
- **Catalytic Converter**
- Catalytic converters, having expensive metals, **platinum-palladium** and **rhodium** as the catalysts, are fitted into automobiles for reducing the emission of poisonous gases.
- Motor vehicles equipped with catalytic converters should use **unleaded petrol** because the lead in the petrol inactivates the catalyst.
- **Working**
- As the exhaust passes through the catalytic converter, **unburnt hydrocarbons** are converted into **carbon dioxide and water**, and carbon monoxide and nitric oxide are changed to carbon dioxide and nitrogen gas, respectively.

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- **Flue-Gas Desulfurization (FGD) systems**
- FGD systems may involve wet scrubbing or dry scrubbing to remove acidic gases, particularly sulphur dioxide (SO_2) and HCl , from flue gases (gases flowing out of vertical exhaust pipes). **Lime** is used in both.
- **Wet FGD**
 - Flue gases are brought in contact with a **liquid or solid slurry absorbent**.
 - The sulphur dioxide dissolves in or reacts with the absorbent & becomes trapped in as wet calcium sulphite.
 - The sulphite can be converted to **gypsum** (used in cement production).
- In **dry FGD** scrubbing, lime is injected directly into flue gas to remove SO_2 and HCl from low-sulphur fuels.

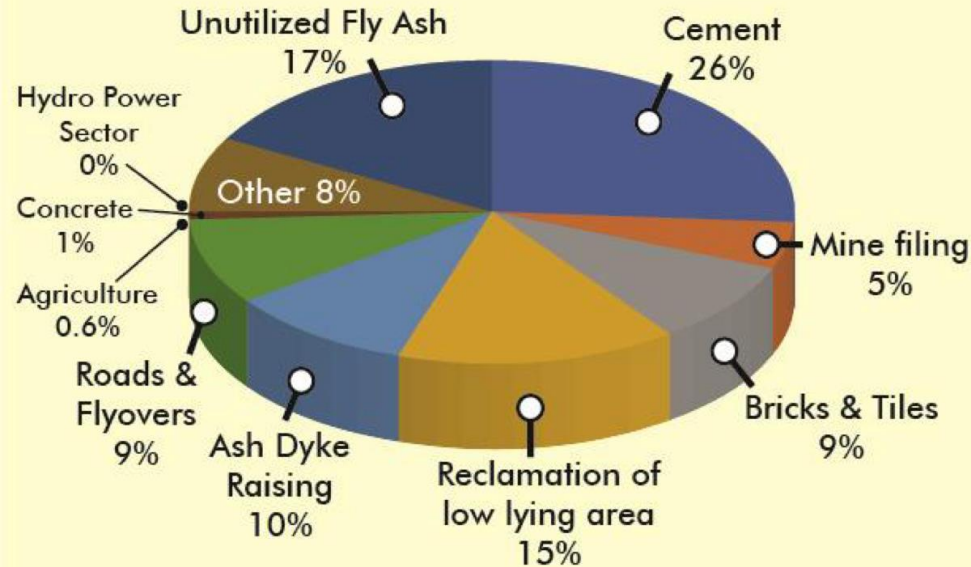
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- **Fly ash**
- Fly ash is a by-product of **coal based power generation**.
- It is a fine powder with substantial amounts of oxides of silica, aluminium and calcium.
- It also contains traces of **Arsenic, Boron, Chromium, Lead etc.** which leads to air and water pollution if disposed on land.
- With **low grade of Indian coal**, its ash content is as **high** as 30-45% in comparison to imported coal with 10-15%.
- With nearly 55% of our total power production through coal and lignite based Thermal Power Plants (TPP), the fly ash generation in India is very high.
- About 83% of Fly Ash is utilized.

- **Recent news**
- The Ministry of Environment, Forests and Climate Change (MoEFCC) has extended fly ash utilization deadline for thermal power plants with the introduction of penalties for noncompliance.
- **Shorter Fly-ash utilization cycle:** Existing provisions allow TPPs to fully utilize fly ash in a four-year cycle in a staggered manner. The new policy will follow a **three-year cycle** for **100% utilization** of Fly-ash with a grace period of a year if the percentage of ash utilization is between 60-80% and two years if it is below 60%.
- Introduction of **Polluter Pays Principle:** A fine of Rs 1,000 per tonne of unutilized ash has been introduced if the plant does not achieve at least 80% ash utilization annually or in three years.

MAJOR MODES OF ASH UTILIZATION DURING 2019-20



- **Acid Rain**
- Acid rain refers to **any precipitation** (rain, fog, mist, snow) **more acidic** than usual ($\text{pH} < 5.6$; $\text{pH} < 7$ is acidic).
- **The pH scale**
- The pH scale measures how acidic or basic (alkaline) a solution is. It ranges from **0 to 14**.
- A pH of 7 is neutral. A pH less than 7 is acidic, and a pH greater than 7 is basic.
- A solution with pH 4 is ten times more acidic than pH 5, and a hundred times more acidic than pH 6.
- pH is based on hydrogen ion concentration in an aqueous solution. pH values decrease as hydrogen ion levels increase. Whilst the pH range is usually 0 to 14, lower and higher values are theoretically possible.

Gas	Source
SO _x (Sulphur Oxides)	Fossil fuel burning, thermal power plants, smelting of metal sulphide ores, industrial sources, industrial production of sulfuric acid in metallurgical, chemical and fertiliser industries, volcanoes, seas and oceans, decomposition of organic matter, etc.
NO _x (Nitrogen Oxides – NO, NO ₂ and N ₂ O)	Fossil fuel burning, lightning, biomass burning, forest fires, oceans, and power plants

- Nitrogen will only react with oxygen at **high temperatures and pressures in lightning bolts and combustion reactions** in power plants or internal combustion engines. Nitric oxide (NO) and nitrogen dioxide (NO₂) are formed under these conditions. Eventually, nitrogen dioxide may react with water in the rain to form nitric acid, **HNO₃**.
- The nitrates thus formed may be utilised by plants as a nutrient (so, the soil gets nitrogen from acid rain).”

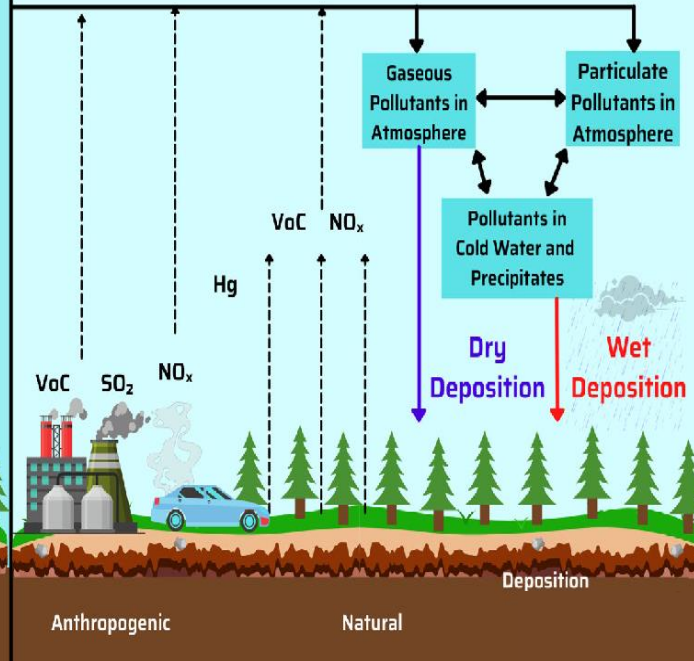
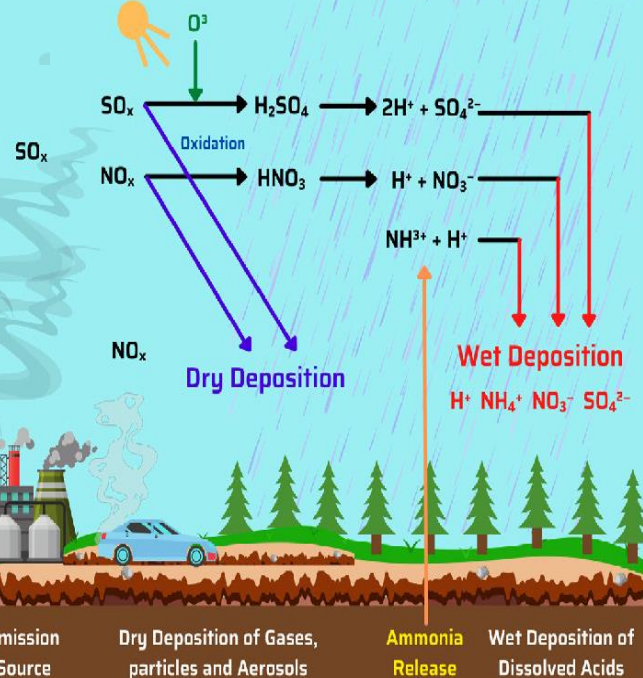
Chemistry of Acid Rain

Basic steps involved in the formation of acid rain:

1. The atmosphere receives **oxides of sulphur and nitrogen** from natural and human-made sources.
2. Some of these oxides fall back directly to the ground as **dry deposition**.
3. Sunlight stimulates the formation of photo oxidants (such as **ozone**) in the atmosphere.
4. These photo oxidants interact with the oxides of sulphur & nitrogen and other gases (like NH_3) to produce **H_2SO_4 (sulphuric acid) and HNO_3 (nitric acid)** by oxidation.
5. Acid rain containing ions of sulphate, nitrate, ammonium and hydrogen falls as **wet deposition**.

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Chemistry of Acid Rain



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Wet Deposition	Dry Deposition
If the acids in the air are blown into areas where the weather is wet, they can fall to the ground as rain, snow, fog, or mist, settling on vegetation as acid depositions.	In areas where the weather is dry, the acid chemicals may become incorporated into dust or smoke and fall to the ground through dry deposition, sticking to the ground, buildings, vegetation, cars, etc.
	About half of the acidity in the atmosphere falls back to earth through dry deposition.

Harmful Effects of Acid Rain

- Effects on soil
- The exchange between hydrogen ions and the nutrient cations like **potassium & magnesium** in the soil cause leaching of the nutrients, making the **soil infertile**.
- **An increase in ammonia** in the soil due to a **decrease** in other nutrients decreases the **decomposition rate**, and the nitrate level of the soil is also found to **decrease**
- Effects on humans
- Bad smell, reduced visibility, and irritation of the skin, eyes and the respiratory tract. Some direct effects include chronic bronchitis, pulmonary emphysema and cancer.

- **Effects on aquatic life:** Eggs or sperms of fish, frogs and other aquatic organisms are sensitive to pH changes. Acid rain kills their gametes affecting the life cycles (**ecosystem imbalances**).
- **Effect on terrestrial life:** Acid rain damages the cuticle of plant leaves and reduces photosynthesis.
- **Effects on microorganisms:** The optimum pH of most bacteria and protozoa is near neutrality. Most **fungi** prefer an **acidic environment**. **Blue-green bacteria prefer an alkaline environment**.
- So, microbial species in the soil and water shift from bacteria-bound to fungi-bound > **delay in the decomposition** of soil organic material
- Effect on buildings, monuments and materials: Limestone and marble are destroyed by acid rain. Ex- Taj Mahal (**Marble cancer**)

- **Acid Rain Areas**
- They are concentrated in the **industrialised** belt of the **northern hemisphere**-Scandinavia, Canada, Japan, the Northeast United States and North-western Europe.
- In India, the first acid rain occurred in **Bombay in 1974**. Instances of acid rain are being reported in many metropolitan cities. Low soil pH is reported from north-eastern India, coastal Karnataka and Kerala, parts of Orissa, West Bengal and Bihar.
- **Acid Rain Control Measures**
- Using **low sulphur fuel or washed coal** (chemical washing of pulverised coal) in thermal plants.
- Buffering: adding a neutralising agent to the acidified water to increase the pH. Usually, lime in the form of calcium oxide & calcium carbonate is used.

- **Plastic Waste Data by Central Pollution Control Board (CPCB)**
- Maharashtra, Tamil Nadu, Gujarat among top Indian Plastic waste producers
- India ranks **5th in generation of plastic waste** with an annual discharge of 3.5 million tonne in 2020.
- Almost **83%** of this comes from just **10 states/ UTs**.
- On a per capita basis, India's plastic waste generation has almost **doubled** over FY2016-20; however, remains lower than most of the developed economies.

- **Plastic Waste Management Rules, 2016**
- Government of India has notified the 2016 rules in place of earlier Plastic Waste (Management and Handling) Rules, 2011 (notified under **Environment Protection Act, 1986**)
- It extended the rules to **all villages**, which were earlier admissible up to municipal areas.
- The 2016 rules increased the minimum thickness of plastic carry bags from 40 to 50 microns & stipulated minimum thickness of **50 microns** for plastic sheets.
- Sachets made of plastic material used for storing, packing, or selling gutkha, tobacco or pan masala are prohibited.

- **Extended Producer's Responsibility (EPR)**
- EPR pins **responsibility** on **producers**, generators & brand owners in waste management and collect back systems.
- Plastic waste management **fee collection** through pre-registration of the producers, importers of plastic carry bags/multilayered packaging and vendors selling the same for establishing the waste management system.
- **Responsibility**
- The **local bodies** shall be responsible for setting up, operationalisation & coordination of the waste management system & for performing associated functions.
- Retailers or street vendors shall not sell, or provide commodities in plastic sheet or multilayered packaging, or they are liable to pay such fines.

- **Reuse of plastic waste**
- Exploring options for the reuse of plastic in various applications, namely, **road construction**, waste-to-oil, and **waste-to-energy**, which will enhance plastic recycling.
- Phasing out non-recyclable multilayered plastic.
- **2018 Amendment**
- The amended rules lay down that the phasing out of **Multilayered Plastic (MLP)** is now applicable only to MLPs which are “non-recyclable, or non-energy recoverable, or with no alternate use”.
- The amended Rules also prescribed an **automated central registration** system to be evolved by the **CPCB** for the registration of the producer/importer/brand owner.
- It omitted Rule 15 i.e. explicit pricing of carry bags mentioned in the 2016 rule.

- **Plastic Waste Management (PWM) Amendment Rules, 2021**
- **Context:**
- Ministry of Environment, Forest, and Climate Change (MoEF&CC) has notified the Plastic Waste Management Amendment Rules, 2021, which prohibits identified **single use plastic** items which have low utility and high littering potential by **2022**.
- India has defined SUP as “a plastic commodity intended to be **used once** for the same purpose before being disposed of or recycled” in its PWM Amendment Rules, 2021.
- These include plastic bags, straws, coffee stirrers, soda and water bottles etc

- The assessment of SUP was conducted by comparing two pillars — the **utility index** of a particular type of SUP and the **environmental impact** of the same.
- The product that scores **low on utility** and **high on environmental impact** should be considered for immediate phase out.
- **Key Provisions**
- **Prohibition** on Manufacture, import, stocking, distribution, sale and use of single-use plastic, including polystyrene and expanded polystyrene, commodities with effect from **1st July 2022**.
- The ban will **not** apply to commodities made of **compostable plastic**.

- Thickness of plastic carry bags increased from 50 microns to 75 microns with effect from 30th September 2021 and to **120 microns** with effect from the **31st December, 2022** [The thickness has been increased so that bags becomes expensive as well as can be reused]
- **Extended Producer Responsibility (EPR):** Plastic packaging waste not covered under present notification shall be collected and managed in an environmentally sustainable way through EPR of the producer, importer and brand owner (PIBO) as per PWM Rules, 2016.
- EPR Guidelines have been given legal force through PWM Amendment Rules, 2021.

- **Implementing agency:** Along with **state pollution bodies, Central Pollution Control Board (CPCB)** will monitor the ban, identify violations, and impose penalties already prescribed under the Environmental Protection Act, 1986.
- **Task Force:** States and UTs had constituted the special task force for elimination of SUP and effective implementation of the PWM Rules, 2016.
- Environment Ministry has also set up a **national level task force** for making coordinated efforts in this direction.
- State /UT Governments and concerned Central Ministries/Departments have also been requested to develop a **comprehensive action plan** for elimination of SUP and its implementation in a **time bound manner**.

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SUP ITEMS BANNED UNDER PWM AMENDMENT RULES, 2016



PLASTIC STICKS

Earbuds with plastic sticks, plastic sticks for balloons, plastic flags, candy sticks, ice-cream sticks.



PACKING/WRAPPING FILMS

Wrapping or packing films around sweet boxes, invitation cards, and cigarette packets.



CUTLERY ITEMS

Plates, cups, glasses, cutlery such as forks, spoons, knives, straw, trays



OTHER ITEMS

Plastic or PVC banners less than 100 microns, stirrers, polystyrene (Thermocol) for decoration.

Steps taken by the Government to ensure enforcement of the ban

Mascot	• PRAKRITI mascot to spread greater awareness among masses and bring behavioural change.
E-governance portals & apps	• National Dashboard on Elimination of Single Use Plastic and Plastic Waste Management set up by MoEFCC to bring all stakeholders at one place and track the progress. • Central Pollution Control Board (CPCB) launched- ○ Extended Producer Responsibility (EPR) Portal for Plastic Packaging for compliance to EPR Obligations by Producers, Importers and Brand-owners. ○ Mobile App for SUP Grievance Redressal to empower citizens to check sale/usage/ manufacturing of SUP in their area. ○ Monitoring module for SUP for local bodies, State Pollution Control Boards (SPCBs)/ Pollution Control Committees (PCCs) and CPCB, to inventorize details of SUP production/ sale & usage in commercial establishments at district level, and on-ground enforcement of ban on SUP.
Directions issued at national, state and local level	• All leading petrochemical industries to not supply plastic raw materials to the industries engaged in banned SUP production. • SPCB/PCCs to modify/revoke consent to operate issued under Air/Water Act to industries engaged in banned SUP production. • Customs Authority to stop the import of banned SUP items. • Local authorities to issue fresh commercial licenses with the condition that SUP items will not be sold on their premises.

Microplastics

- Microplastics are shreds of plastic **less than 5 mm in length** but **larger than 1 micrometre**.
- Microplastics are used in cosmetics, personal care products, industrial scrubbers, microfibers in textiles and virgin resin pellets used in plastic manufacturing processes.
- Several studies have established the presence of microplastics in groundwater. A study estimated that the average human ends up consuming at least 50,000 microplastics in food every year. Their impact on tiny marine organisms is even significant since their injection leaves them starved and affects the marine food web.

74 tonnes of microplastics fell from the air on Auckland in 2020

Scientists have been discussing the possibility that microplastic mists and clouds exist in the atmosphere but this is the first study to quantify the magnitude of the problem

- According to a study, we **may inhale substantial quantities of microplastics with every breath.** And these plastics will eventually flow into our blood and accumulate in our organs.
- Scientists have been discussing the possibility that microplastic mists and clouds exist in the atmosphere but this is the first study to quantify the magnitude of the problem.
- **Read here:** <https://www.downtoearth.org.in/video/pollution/74-tonnes-of-microplastics-fell-from-the-air-on-auckland-in-2020-86600>

Microbeads

- Microbeads ($>0.1 \mu\text{m}$ and $< 5 \text{ mm}$) are very **tiny pieces of plastic** that are added to health and beauty products, such as in some cleansers and toothpaste (cooling crystals).
- However, one of the **main contributors to microbeads pollution** is not manufactured microbeads, but regular plastic waste, 90% of which are not recycled.

Polyethylene Terephthalate (PET Plastic)

- **PET (polyethylene terephthalate)**, the chemical name for polyester. It is a clear, strong, lightweight plastic widely used for packaging foods and beverages, especially convenience-sized soft drinks, juices, water, cooking oils, etc.
- PET is **entirely recyclable**. They can easily be identified by the #1 in the triangular "chasing arrows" code, which is usually moulded into the bottom or side of the container. No other plastic carries the #1 code.



- **Status of e-waste in India**
- **3rd largest Electronic waste** producer in the world after China and United States.
- Produces about 4 mMT (million Metric Tonnes) of e-waste.
- This is expected to increase **40-fold by 2050**.
- **Computer devices** accounts for nearly **70%** of annual e-waste production, followed by telecom sector, medical equipment and electric equipment.
- Only **1.5%** of electronic waste generated is recycled through institutional processes.
- Uttar Pradesh, Uttarakhand, Tamil Nadu, and Haryana are among the States that have a bigger capacity to dismantle and recycle e-waste.

- **E-waste (Management) Rules 2022**
- **Context:** Ministry of Environment, Forest and Climate Change (MoEF&CC) has notified E-waste (management) Rules 2022, in the exercise of the powers conferred by the **Environment (Protection) Act, 1986**. [Recall our discussion that EPA is an “umbrella” legislation]
- **Key Provisions:**
- 'E-waste' **Definition:** Electrical and electronic equipment, whole or in part discarded as waste, and rejects from manufacturing, refurbishment and repair processes

- **Coverage:** Every manufacturer, producer, refurbisher, dismantler, and recycler involved processing of e-waste with certain exceptions
- **Compulsory Registration** of manufacturer, producer, refurbisher, or recycler of the e-products with Central Pollution Control Board (CPCB).
- Introduction of **Extended Producer Responsibility (EPR) Certificates** to incentivize registered electronic waste recyclers by introducing (not part of 2016 Rules).
- CPCB to conduct **random sampling** of electrical and electronic equipment placed on the market to **monitor and verify** the compliance of reduction of hazardous substances provisions.

- **The Right To Repair Movement**
- Right to Repair electronic products is a reference to the need for government legislation that is intended to allow consumers the ability to repair and modify their own electronic devices.
- The movement traces its roots back to 1950s.
- The concept originated from USA from the automotive industry.
- In July 2022, the **United States passed the Fair Repair Act, 2022**, on the right to repair.
- The United Kingdom and European Union passed legislation such as **Right to Repair Regulations and Right to Repair** regarding this right.
- Recently, in India, the Department of Consumer Affairs announced the formation of a **committee** under the chairmanship of **Nidhi Khare** to develop a **comprehensive framework** for the right to repair.

- **Solid Waste Management Rules (2016)**
- These replace the Municipal Solid Wastes (Management and Handling) Rules, 2000. (notified under **Environment Protection Act, 1986**)
- These Rules are now applicable **beyond Municipal areas** and extend to urban agglomerations, notified industrial townships, areas under the control of Indian Railways, airports, etc., defence establishments, places of pilgrims, etc.
- Responsibilities of Generators have been introduced to segregate waste into **3 streams** before handing it over to the collector:
 - 1) **Wet** (Biodegradable)
 - 2) **Dry** (Plastic, Paper, metal, wood, etc.)
 - 3) Domestic **hazardous wastes** (diapers, napkins, containers of cleaning agents, mosquito repellents, etc.)

- **Segregation at Source**
- **Source segregation** of waste has been mandated to channel waste to wealth by **recovery, reuse and recycling**
- Hotels and restaurants will also be required to segregate biodegradable waste & set up a system of collection to ensure that such food waste is utilised for composting/biomethanation.
- All resident welfare and market associations and gated communities with an area of **above 5,000 sq m** will have to **segregate waste at the source**. They have to hand over recyclable material to authorised pickers and recyclers or the urban local body.

- **Collect Back scheme for packaging waste**
- **Brand owners** who sell or market their products in non-biodegradable packaging materials should have a system to collect back the packaging waste generated due to their production.
- **User Fees for Collection**
- **Municipal authorities** will **levy user** fees for collection, disposal and processing from bulk generators.
- As per the rules, the generator will have to pay a “**User Fee**” to the waste collector & a “**Spot Fine**” for littering & non-segregation, the quantum of which will be decided by the local bodies.

- **Waste Processing & Treatment**
- Bio-degradable waste should be processed and disposed of through **composting/biomethanation**.
- Rules have mandated **bioremediation** or capping of old and abandoned dump sites within 5 years. Waste processing facilities will have to be set up by local bodies within the stipulated time frame.
- **Promotion of Waste to Energy**
- All industrial units **within 100 km** of a solid waste-based **Refuse-Derived Fuel (RDF) Plant** must make arrangements to replace at least 5 per cent of their fuel requirement with RDF so produced.
- Ministry of New and Renewable Energy should facilitate infrastructure creation for Waste to Energy plants and provide appropriate subsidies or incentives for such Waste-to-Energy plants.

- **Revision of Parameters**
- Landfill site shall be 100 meters away from a river, 200 meters from a pond, 500 meters away from highways, habitations, public parks and water supply wells and 20 km away from airports/airbases.
- **Emission standards** are completely amended for dioxins, furans, particulate matter, etc.
- The compost standards have been amended to align with Fertilizer Control Order.
- **Constitution of Central Monitoring Committee**
- Central Monitoring Committee under the chairmanship of the **Secretary, MoEFCC**, to monitor the implementation of the rules.

- Treatment of solid waste
- As per 12th schedule and 74th Amendment act, ULB (urban local bodies) are responsible for keeping cities clean.
- Incineration plants
- The process of **burning waste** in large furnaces at high temperatures is known as **incineration**.
- In these plants, the recyclable material is segregated, and the rest is burnt.
- Burning garbage is not a clean process, as it produces tons of **toxic ash** and **pollutes** the air and water.
- At present, incineration is kept as a **last resort** and is used mainly for treating infectious waste.

- **Pyrolysis**
- Pyrolysis is a process of **combustion (burning)** of material in the **absence of oxygen**, or under a controlled atmosphere of oxygen.
- It is an **alternative** to incineration.
- Pyrolysis of carbonaceous wastes like firewood, coconut, palm waste, corn combs, cashew shells, rice husk paddy straw and sawdust, yields **charcoal** and products like tar, methyl alcohol, acetic acid, acetone and fuel gas.

- **Plasma gasification**
- Plasma gasification is an **extreme thermal process** (uses a combination of electricity and high temperatures) using **plasma** (without combustion) which converts **organic matter** into a **syngas** (synthesis gas - made up of hydrogen and carbon monoxide).
- The **inorganic** wastes are converted into **slag**, a solid waste.
- Plasma gasification is a **cleaner alternative** to landfills, reducing or eliminating toxicity while avoiding the landfilling of huge amounts of garbage

- **Bio-Medical Waste (Management and Handling) Rules, 2016**
- Bio-Medical Waste Management Rules 2016 is an improvement to the 1998 rules. (notified under **Environment Protection Act, 1986**)
- Biomedical waste comprises **human & animal anatomical waste** and treatment apparatus like needles used in health care facilities (HCF – hospitals, laboratories, immunisation programmes, etc.)
- **Salient features:**
- Phase out **chlorinated plastic bags**, gloves and blood bags within two years.
- **Pre-treatment** of the laboratory waste, microbiological waste, and blood samples through sterilisation onsite.

- Establish a **bar-code system** for bags or containers containing bio-medical waste for disposal.
- Bio-medical waste has been classified into **4 categories** instead earlier ten to improve segregation at source.
- **State Government** to provide land for **common bio-medical waste treatment and disposal facilities**.
- No occupier shall establish an on-site treatment and disposal facility if a service of it is available at a distance of **seventy-five kilometers**
- Operator of a common bio-medical waste treatment and disposal facility to ensure the **timely collection** of bio-medical waste from the HCFs.
- **Vaccination camps**, blood donation camps, etc., will come under the ambit of these rules.

- **Procedure to be followed**
- The **hospitals** must put in place the mechanisms for effective disposal either directly or through common biomedical waste treatment and disposal facilities.
- The hospitals servicing 1000 patients or more per month must obtain authorisation and segregate biomedical waste into categories as specified by the rules.
- For example, syringes, needles and blood-soiled bandages should be all disposed of in a red-coloured bag or bin, where they will later be incinerated (destroy by burning).
- If **body fluids** are present, the material needs to be **incinerated**.

Pollution

Cat.	Type of Bag/ Container used	Type of Waste	Treatment/Disposal options
Yellow 	non-chlorinated plastic bags Separate collection system leading to effluent treatment system	a) Human Anatomical Waste b) Animal Anatomical Waste c) Soiled Waste d) Expired or Discarded Medicines e) Chemical Waste f) Micro, Bio-t and other clinical lab waste g) Chemical Liquid Waste	Incineration or Plasma pyrolysis or deep burial*
Red 	non-chlorinated plastic bags or containers	Contaminated Waste (Recyclable) tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles) and gloves	Autoclaving/microwaving /hydroclaving and then sent for recycling not be sent to landfill
White 	(Translucent) Puncture, Leak, tamper proof containers	Waste sharps including Metals	Auto or Dry Heat Sterilization followed by shredding or mutation or encapsulation
Blue 	Cardboard boxes with blue colored marking	Glassware	Disinfection or autoclaving, microwaving, hydroclaving and then sent for recycling

- **Hazardous Wastes (Management and Handling) Rules, 2016**
- It has been notified under **Environment Protection Act, 1986**.
- For the first time, Rules have been made to distinguish between **hazardous Waste & other wastes**.
- **Other wastes** include waste tyres, paper waste, metal scrap, used electronic items, etc. and are recognised as a **resource** for recycling and reuse.
- **Salient Features**
- The ambit of the Rules has been expanded by including ‘Other Waste’.
- Waste Management hierarchy in the sequence of priority of prevention, minimisation, reuse, recycling, recovery, co-processing; and safe disposal has been incorporated

- The rules encourage the **reduction of hazardous waste generation** and encourage its recycling and reusing
- They specify strict guidelines related to the import/export or even storage and transportation of hazardous wastes.
- Import & Export of Hazardous Waste
- **No country** can **export** hazardous waste to India for final disposal. This means that India only imports hazardous wastes to recycle, reuse or for other utilisation.
- The rules specify the procedure for importing and exporting hazardous waste to and from India.

- Treatment, Storage and Disposal Facility for Hazardous Wastes
- 2016 rules give a clear direction of how the facility for treatment, storage & disposal is to be established.
- **Permission** from the **SPCB** is required for the layout in this regard.
- **Packaging, Labelling and Transport of Hazardous and Other Wastes**
- **CPCB** provides extensive guidelines for packaging and labelling. If the waste is to be transported to a facility in a different state for its final disposal, a 'No Objection Certificate' is required on the sender's part from the SPCBs of both states.

- **2019 Amendment**
- Amendment has been made considering the “**Ease of Doing Business**” & boosting the “**Make in India**” initiative by simplifying the procedures under the Rules, while simultaneously upholding the principles of sustainable development & ensuring minimal impact on the environment.
- **Salient Features**
- Solid plastic waste has been prohibited from being imported into the country, including in Special Economic Zones (SEZ) and by Export Oriented Units (EOU).
- Exporters of **silk waste** have now been given an **exemption** from requiring permission from MoEFCC.

- **Electrical and electronic assemblies** and components manufactured in and exported from India, if found **defective**, can now be imported back into the country without obtaining permission from MoEFCC.
- **Industries** that do not require **consent** under the Water (Prevention and Control of Pollution) Act 1974 and the Air (Prevention and Control of Pollution) Act 1981, are now **exempted** from requiring authorization also under the Hazardous and Other Wastes Rules, 2016.

Measuring Pollution Load in Water Dissolved Oxygen (DO)

- Optimum DO content in water is important for the survival of aquatic organisms.
- The presence of organic and inorganic wastes decreases the DO content due to the **high decomposition rate and O₂ consumption**.
- Several factors, such as surface turbulence, photosynthetic activity, O₂ consumption by organisms and decomposition of organic matter, determine the amount of DO in water.
- Water having DO content below **8.0 mg/L** may be considered contaminated, and below **4.0 mg/L** is considered to be highly polluted.

Biological Oxygen Demand (BOD)

- Water pollution by organic wastes is measured in terms of Biochemical Oxygen Demand (BOD).
- BOD is the **amount of dissolved oxygen needed by bacteria to decompose the organic wastes present in water**. It is expressed in milligrams of oxygen per litre of water.
- The **higher value of BOD indicates high pollution** due to biodegradable organic wastes and low DO content of water. Since BOD is limited to biodegradable materials, it is not a reliable method of measuring water pollution.

Pollution



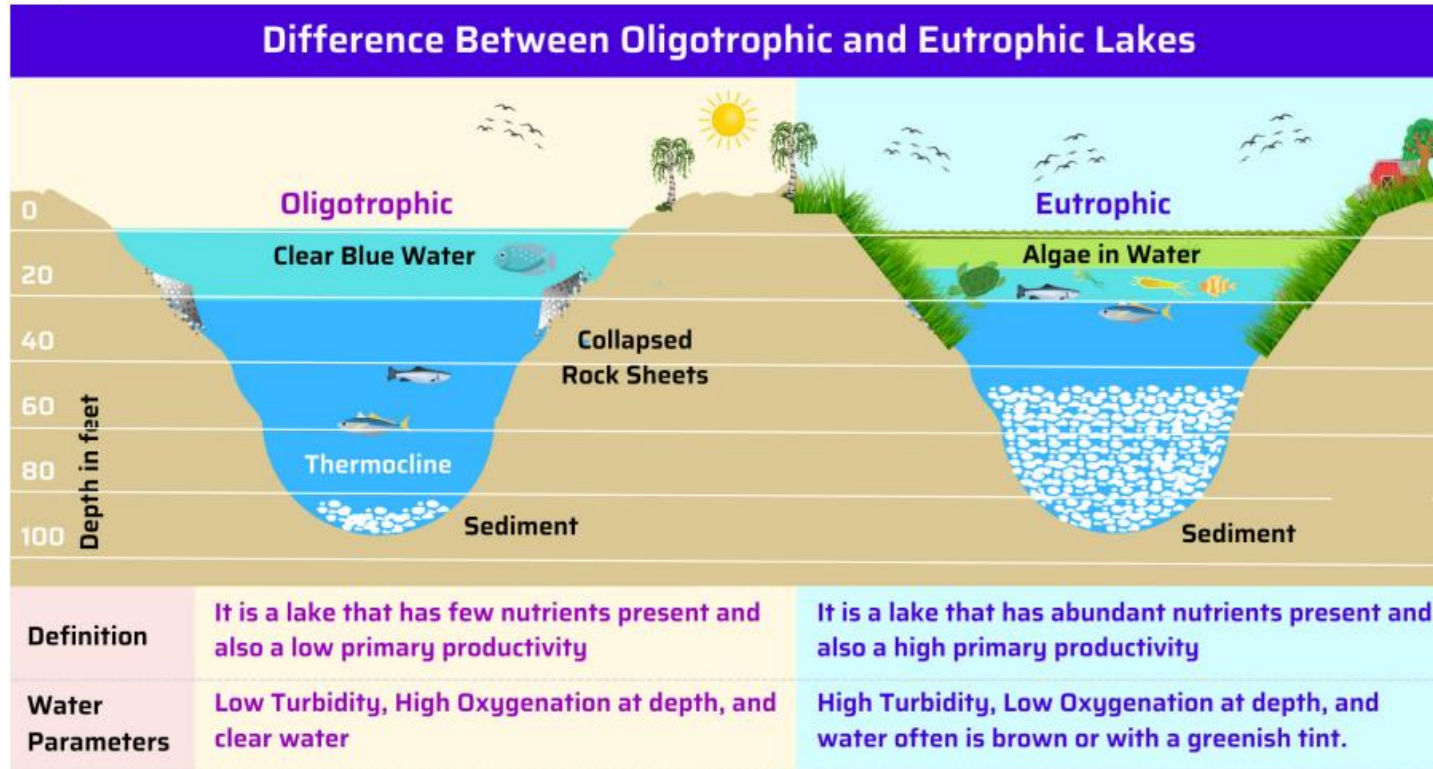
BOD Level in mg/liter	Water Quality
1 - 2	Very Good: There will not be much organic matter present in the water supply.
3 - 5	Fair: Moderately Clean
6 - 9	Poor: Somewhat Polluted - Usually indicates that organic matter present and microorganisms are decomposing that waste.
100 or more	Very Poor: Very Polluted - Contains organic matter.

Chemical Oxygen Demand (COD)

- Chemical oxygen demand (COD) is a slightly better mode used to measure pollution load in the water.
- It measures **the amount of oxygen in parts per million required to oxidise organic** (biodegradable and non-biodegradable) and oxidisable inorganic compounds in the water sample.

Eutrophication and Ageing of Lakes

- The nutrient enrichment of the lakes promotes the growth of algae, aquatic plants, and various fauna. This process is known as **natural eutrophication**.
- Human activities cause similar nutrient enrichment of lakes at an accelerated rate, and the consequent ageing phenomenon is known as **cultural eutrophication**.
- Based on their nutrient content, lakes are categorised as **Oligotrophic (very low in nutrients)**, **Mesotrophic (moderate nutrients)** and **Eutrophic (highly nutrient-rich)**. Most lakes in India are either eutrophic or mesotrophic because of the nutrients derived from their surroundings or the organic wastes entering them.



- **Eutrophication effects :**
- Collapse of food chains
- New Species Invasion
- Loss of freshwater lakes
- Loss of coral reefs

Eutrophication and Algal Blooms

- Eutrophication is the **response to the addition of nutrients such as nitrates and phosphates naturally or artificially, fertilising the aquatic ecosystem.**
- **Phytoplankton** thrives on the excess nutrients, and their population explosion covers almost the entire surface layer, restricting the penetration of sunlight. This condition is known as an **algal bloom.**
- It results in the **death of aquatic plants and hence restricts the replenishment of oxygen.**

Mechanism

- **Phytoplankton are photosynthetic during the daytime, adding oxygen to the aquatic ecosystem. But during the night, they consume far more oxygen as they respire aggressively.** Therefore, algal blooms accentuate the rate of oxygen depletion as the population of phytoplankton is exceedingly high.
- The primary consumers like zooplankton & small fish are killed due to oxygen deprivation caused by algal blooms, adversely affecting the food chain.
- Further, more oxygen is taken up by microorganisms during the decomposition process of dead algae, plants, and fishes. The new anaerobic conditions (absence of oxygen) created promote the growth of bacteria such as **Clostridium botulinum** which produces toxins deadly to aquatic organisms, birds, and mammals.

Harmful Algal Blooms

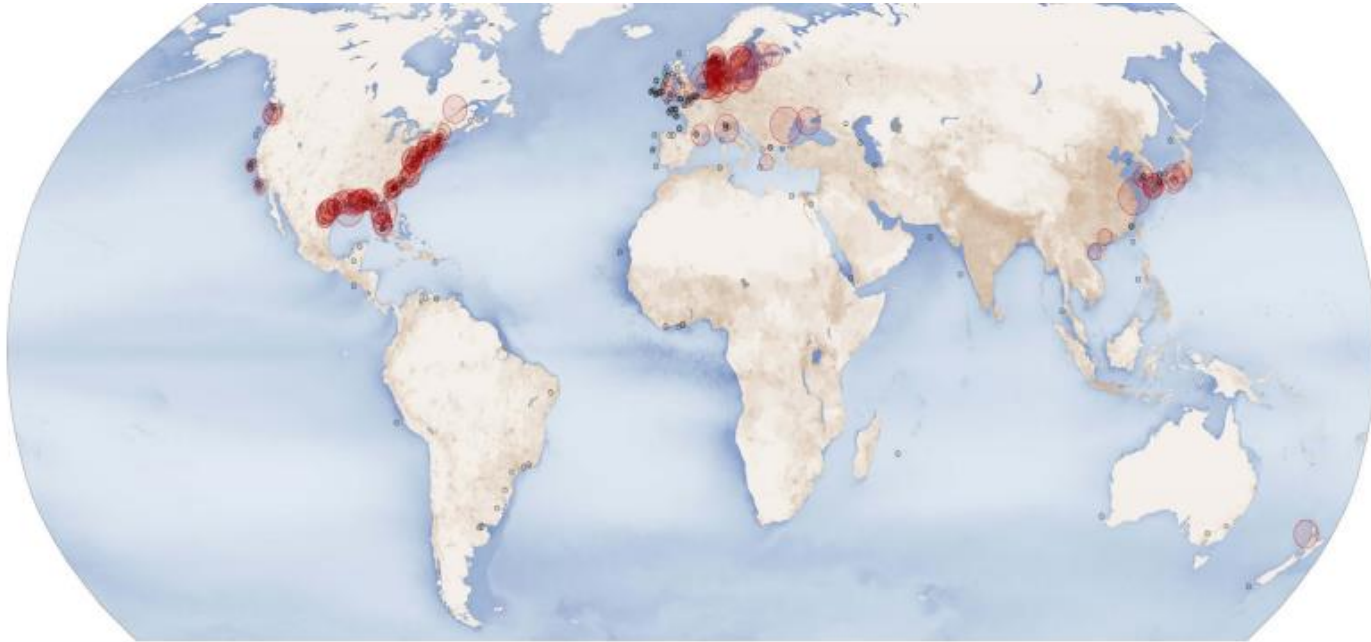
- Most algal blooms are not harmful, **but some produce toxins** (neuro and hepatotoxins), which can kill aquatic organisms (E.g., Shellfish poisoning) and pose a threat to humans and are known as Harmful Algal Blooms (HABs).
- They adversely **affect commercial and recreational fishing, tourism, and valued habitats.**

Dead zones

- Dead zones (biological deserts or hypoxic zones) are areas in the ocean with very low oxygen concentrations (hypoxic conditions). They **emerge when the influx of excess chemical nutrients spurs algae growth (algal blooms)**. These zones usually occur 200-800 meters (in the saltwater layer) below the surface.
- **Hypoxic zones** can occur **naturally** (due to the upwelling of excess nutrients). They can be created or enhanced by human activity to form dead zones. Dead zones are detrimental to animal life. Most of the animal life either dies or migrates from the zone.

- Dead zones are increasing in the coastal delta and estuarine regions. One of the largest dead zones forms in the **Gulf of Mexico** every spring (farmers fertilise their crops, and rain washes fertiliser off the land into rivers). There's a dead zone in the **Gulf of Oman**, and it's growing.

Pollution

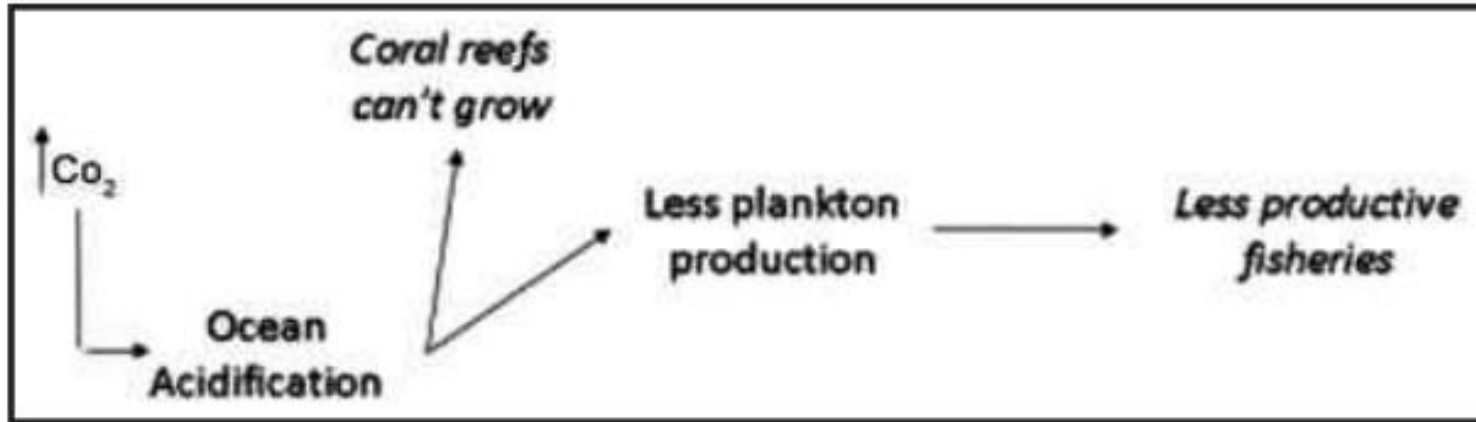


Red circles show the size and location of dead zones in 2010. Dark blue regions indicate overly fertile water that may give rise to dead zones. ([NASA Earth Observatory](#))

Ocean Acidification

- Oceans are an important reservoir of CO_2 : they absorb a significant quantity.
- Ocean acidification is the change in ocean chemistry - lowering of ocean pH (i.e. increase in concentration of hydrogen ions) driven by the uptake of carbon compounds by the ocean from the atmosphere
- As the uptake of atmospheric carbon dioxide by the ocean increases, the concentration of hydrogen ions in the ocean increases, the concentration of carbonate ions decreases, the pH of the oceans decreases and the oceans become less alkaline – this process is known as ocean acidification.

Pollution



Mechanism of Ocean Acidification

- An estimated 30-40% of the carbon dioxide from human activity released into the atmosphere dissolves into oceans, rivers, and lakes. To achieve chemical equilibrium, a part of it reacts with water to form carbonic acid.
- Some carbonic acid molecules react with a water molecule to give a bicarbonate ion and a hydronium ion (H^+), thus increasing ocean acidity (H^+ ion concentration).
- The **pH of the ocean surface waters has decreased by about 0.1 pH unit** (i.e. 26% increase in ocean hydrogen ion concentration) since the beginning of the industrial revolution.

Effects of Ocean Acidification

- **Reduced Buffering Capacity:** Oceans are an essential reservoir of CO₂, absorbing a significant quantity of it (one-third) produced by anthropogenic activities and effectively buffering climate change. The uptake of atmospheric carbon dioxide is occurring at a rate exceeding the natural buffering capacity of the oceans. Increasing acidity depresses metabolic rates and immune responses in some organisms.

Effects on Marine Calcifying Organisms:

- Seawater absorbs CO_2 to **produce carbonic acid, bicarbonate and carbonate ions**. However, the increase in atmospheric CO_2 levels leads to a decrease in pH level and an increase in the concentration of carbonic acid and bicarbonate ions, **causing a decrease in the concentration of carbonate ions**.
- The **decreased amount of carbonate ions** makes it **more difficult for marine calcifying organisms**, such as coral (calcareous corals) and some plankton (calcareous plankton), to form biogenic calcium carbonate.
- This **accentuates coral bleaching**, and commercial fisheries are also threatened as the calcifying organisms form the base of the aquatic food webs.

Increasing ocean acidity will hit the marine food chain, says study

Increasing acidification impacts phytoplankton, which in turn will drastically affect the fish population

Findings:

- Ocean acidification **prompted some species to grow faster and others slower**, it also changed the natural competition between species.
- It **inhibits shell growth** in marine animals and is suspected as **a cause of reproductive disorders** in some fish.

Read:

<https://www.downtoearth.org.in/news/climate-change/increasing-ocean-acidity-will-hit-the-marine-food-chain-says-study-50602>

Polluting Industries

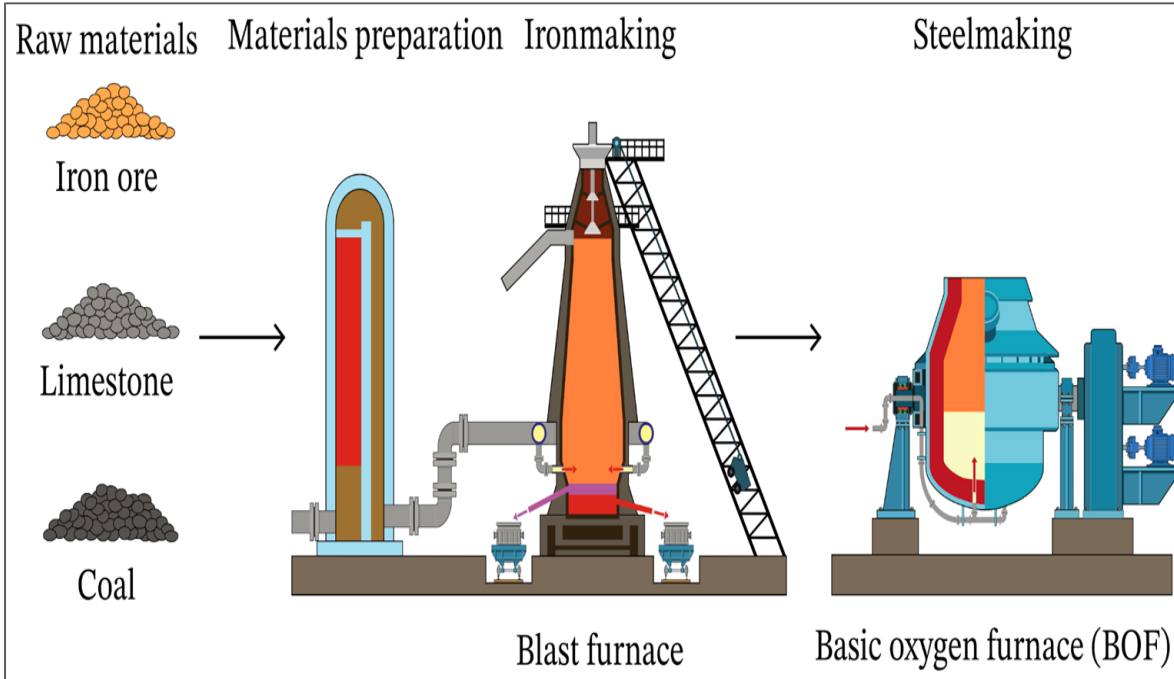
Iron and Steel Industry

- **Processes involved**

- Unwanted impurities are removed by **smelting iron** ore in a blast furnace.
- Major impurities include **sulphur** (which forms iron sulphide, which dramatically reduces the strength of steel), **lead** (improves the machinability of the steel when present in small quantities), **oxygen** (oxides make iron and steel weak), etc.

- In a blast furnace, **fuel** (coke — has far fewer impurities than coal), **iron ore**, and **flux** (limestone — removes sulphur and other impurities into slag) are continuously supplied.
- The byproducts obtained are **liquid slag**, liquid iron (pig iron — an intermediate product of smelting iron ore; has oxides) and gases.
- Oxygen in the iron oxides is reduced by a series of chemical reactions that produce CO and CO₂.
- [Pig iron to steel> Liquid (pig) iron> Cast iron (cooled liquid iron; brittle; carbon content greater than 2%; wrought iron (weak) = liquid iron + slag)> Steel (carbon content is up to 2.1%; it does corrode)> Stainless steel (steel + 10.5% chromium + Nickel, manganese, molybdenum, etc.; it does not corrode).]

Pollution



- **Steel Slag**
- Slag full of impurities such as **calcium sulphide** (CaS) and oxides of silica, alumina, magnesia, calcium (CaO), etc., that entered with the iron ore or coke.
- Only a small percentage of slag goes into landfills.
- **Cement** made with blast furnace slag has lower permeability and is more **durable** than Portland (regular) cement. It is used as an aggregate in concrete, cement clinker, asphalt concrete, asphalt and road bases.
- **Soil improvement:** dissolution of slag generates alkalinity that can be used to precipitate out metals, sulfates, and excess nutrients (like nitrogen, phosphorus, potassium) in wastewater treatment.
- **Soil conditioner:** ferrous slags have been used to rebalance soil pH and as fertilisers (as sources of calcium and magnesium).

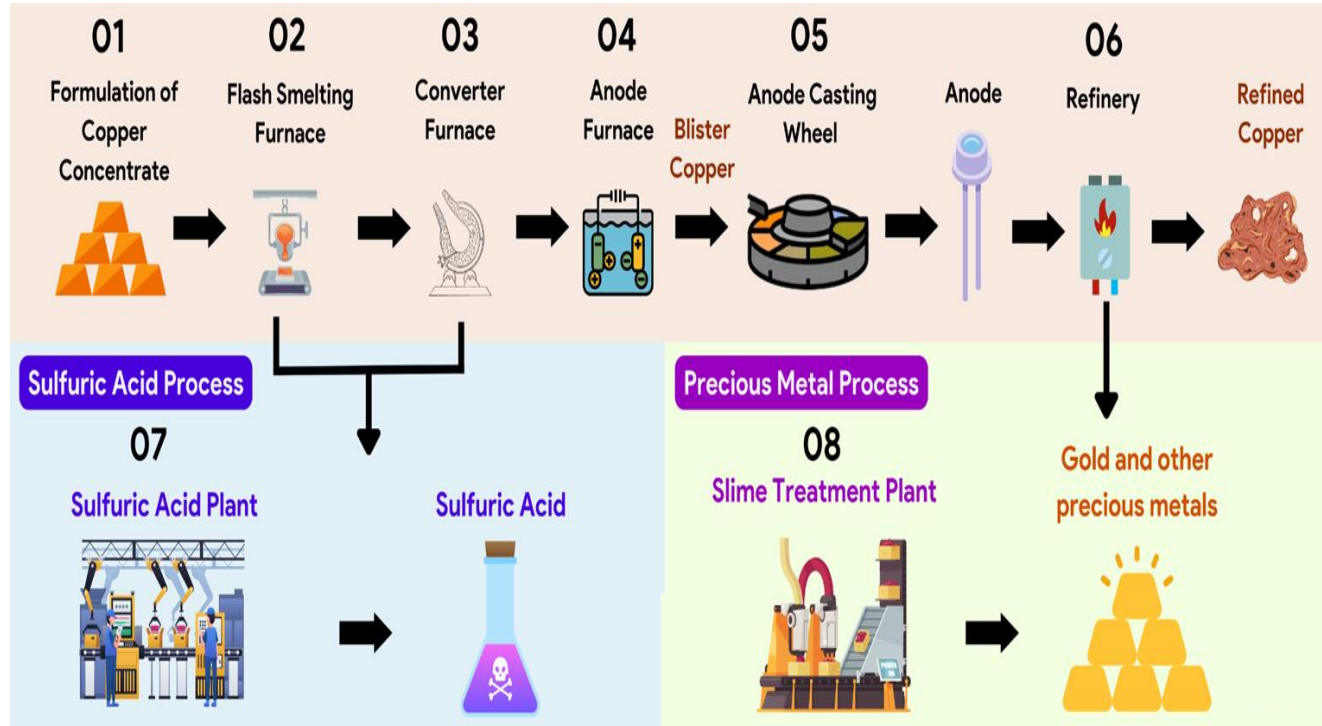
- **Air Pollution**
- The industry burns a lot of coal (thermal power; coke in the blast furnace), causing air pollution in the form of **PM2.5 and PM10**, Carbon Dioxide, **Sulphur Oxides** (sulphur is eliminated as SO₂ in the blast furnace),
- **Nitrogen Oxides**, Carbon Monoxide, Hydrogen Sulphide, Non-Methane Volatile Organic Compounds (**NMVOC**), etc.
- Coke ovens emit naphthalene which is highly toxic and can cause cancer (carcinogenic).

- **Magnetite Pollution**
- Magnetite pollution refers to the presence of a **magnetic mineral** called Magnetite (Fe_3O_4) in the environment, as a result of human activities such as mining, steel production and industrial processes.
- Magnetic particles can interfere with the **migratory patterns of birds** and the operation of electronic equipment, such as **compasses** and navigation systems.

- **Copper Smelting Industry**
- **Processes Involved**
- Mining of **raw chalcopyrite ore** (CuFeS_2 — copper iron sulfide; most abundant copper ore mineral; $<1\%$ Cu).
- Improving chalcopyrite ore concentration (34.5% Cu, 30.5% Fe, and 35.0% S) by crushing, grinding, and **flotation purification** (oils are used as certain metals have an affinity to oils) at the mine site.
- **Roasting** of improved ore concentrate is performed in copper smelters (mostly near ports) to reduce impurities, including sulfur, antimony, arsenic, lead, etc.
 - It eliminates 20-50% sulfur as highly concentrated **sulfur dioxide (SO_2)**. It is converted to concentrated **sulfuric acid** and transported to fertilisers, pharmaceuticals, paper bleaching, petroleum refineries and other industries.

- **Smelting** of roasted ore concentrate produces matte (65% Cu), a molten mixture of copper sulfide (Cu_2S), iron sulfide (FeS), and slag consisting of iron oxide and heavy metals.
- Converting the matter in a converter furnace yields high-grade blister copper (~99% Cu).
- **Blister copper** is refined in an anode furnace, cast into anodes to remove oxygen (99.5% Cu).
- **Electrolytic refining:** Copper from the anode plates is electrolytically deposited on the stainless steel cathode plate, resulting in copper with a grade of 99.99%.
- Slime generated in electrolytic refining contains precious metals such as gold, silver, selenium, tellurium etc.
 - These metals are recovered in a **slime treatment plant**.

Pollution



- **Copper slag**
- Copper slag is mainly used for **abrasive surface blast-cleaning** (used to shape the surface of the metal, stone, concrete, etc.). It can be used in **road construction** and the production of **cement**, mortar, and concrete as raw materials for clinker, coarse and fine aggregates.
- **Pollution**
- Leaching of impurities from copper ore concentrate and slag: **radon** (emitted from the natural radioactive decay of uranium, and copper ores), iron, arsenic, antimony, mercury, lead, cadmium, selenium, magnesium, aluminium, cobalt, tin, nickel, manganese, nitrates, fluorides, etc.
- As most copper ores are sulphur-based, the smelting process releases **Sulphur Dioxide**. When the concentration of sulphur dioxide is too high, the industries convert it into sulphuric acid, an irritant and water contaminant.

- **Petroleum Refining & Petrochemicals**
- The petroleum industry is subdivided into **upstream**, **midstream**, and **downstream** segments.
- The upstream deals with the exploration and mining of crude oil, the midstream includes storage and transport of crude, while the downstream involves refining.
- **Petroleum Refining**
- **Fractional distillation:** separation of different fractions (hydrocarbon compounds) of crude oil based on their boiling point differences.
- **Conversion processes:** breaking down long chain molecules into more valuable smaller ones by heating.
- **Treating:** separating the impurities such as sulfur, nitrogen and heavy metals.

- **Pollution**
- Air pollutants include **particulate matter** (PM), carbon dioxide (CO₂), nitrogen oxides (NO_x), carbon monoxide (CO), **hydrogen sulfide** (H₂S), **sulfur dioxide** (SO₂), natural gas (methane), lead, VOCs including cancer-causing benzene pollutants, etc.
- Refineries use deep-injection into **underground wells** and coastal waters to dispose of wastewater and **oil residues** generated inside the plants, and some of these wastes end up in aquifers and groundwater.

- **Furnace oil** is a residue of crude oil which is **used** for
 - Marine engines and slow speed engines for **power generation**,
 - Drying tea leaves,
 - Gas turbines for power generation etc
- The emissions of **oxides of sulphur** are a direct result of the sulphur content of the fuel oil.

Heavy Metal Contamination

75% of river monitoring stations report heavy metal pollution: Centre for Science and Environment

Heavy toxic metals such as lead, iron, nickel, cadmium, arsenic, chromium and copper were present at alarming levels, according to the NGO

Heavy Metals

- **They are naturally occurring elements**, with high atomic weight and density at least 5 times greater than that of water.
Examples: Lead, iron, nickel, cadmium, arsenic, chromium and copper.
- **Effects of heavy metals exposure:**
 - **Health:** Slowly progressing physical, muscular, and neurological degenerative processes that mimic Alzheimer's, Parkinson's disease etc.
 - **Environment:** Affect biodegradability of organic pollutants, making them less degradable.
 - **Plants:** Influences soil fertility, disturbs photosynthesis etc.

Other common water contaminants and their impact on human health



Arsenic

Black-Foot Disease, painful and disfiguring skin conditions melanosis, keratosis, and hyperkeratosis, liver and kidney disease, and fatalities from heart and lung disease and multiple cancers.



Fluoride

Skeletal fluorosis, dental fluorosis, etc.



Manganese

Impairs the intellectual development etc.



Uranium

Chemical damage to the kidneys, increase in bone cancer risk etc



Nitrate

Blue baby syndrome (methemoglobinemia), increased risks of colorectal cancer, thyroid disease, and neural tube defects etc.

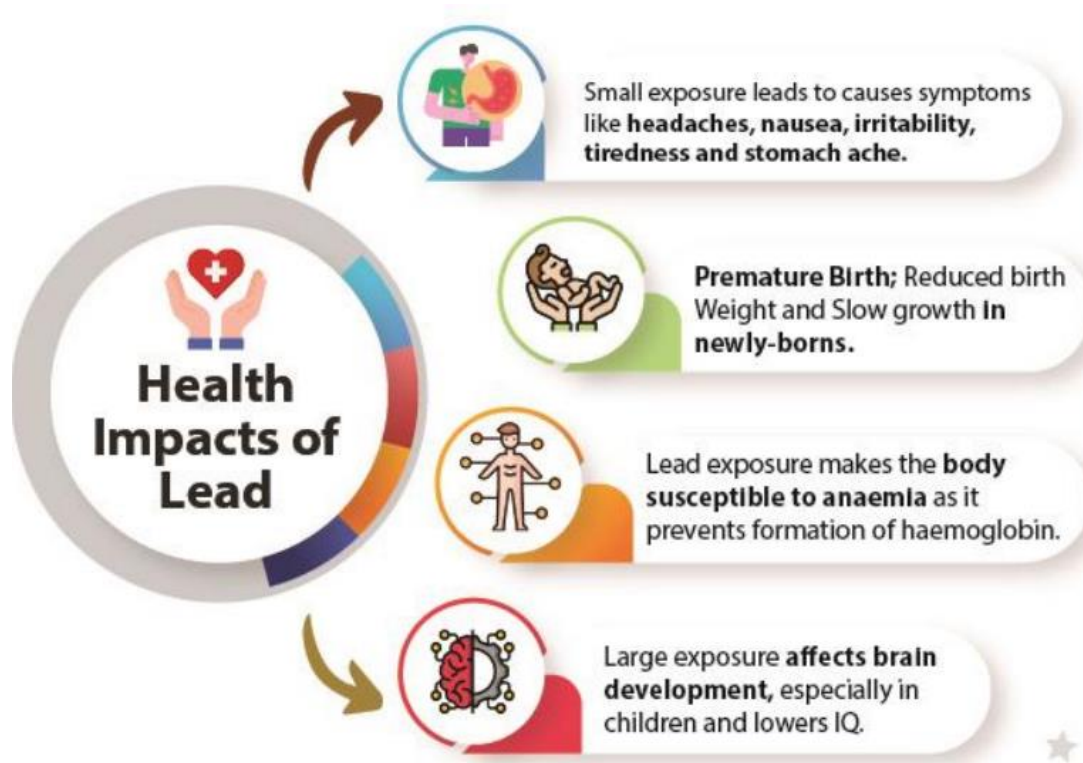


Lead Poisoning

- A report about lead poisoning was prepared jointly by government think tank **Niti Aayog and Council of Scientific & Industrial Research**. It says India bears world's highest health, economic burden due to lead poisoning.

Sources of lead:

- Naturally found in **the Earth's crust**.
- Mining, smelting and **refining industries and their waste**.
- Household Products such as Cosmetic Products, Ayurveda medicines etc.
- Present in **printed circuit boards and batteries** due to unscientific recycling
- Potteries (e.g. utensils and glazed ceramic wares), water bodies etc.



Per And Polyfluoroalkyl Substances (PFAs)

- Chemicals with partially or completely fluorinated carbon chains of varied lengths. They are referred as “**forever chemicals**” as do not degrade easily in the environment due to strong carbon-fluorine bonds.
- PFAS like perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) listed under **Stockholm Convention on Persistent Organic Pollutants** (their production and use are restricted or eliminated in the Parties.)

Common uses of PFAS



Non- stick cook
and bake- ware



Fire- fighting
foams



Water- and oil- proof
apparel



Stain resistant
upholstery carpet,
etc.



Biocides, household
agents such as
cleaning agents and
impregnation sprays.



Food
packing



Concerns associated with use of PFAS:

- Found in the blood of people and animals.
- High exposure associated with health risks like decreased fertility, developmental effects in children, interference with body hormones, increased cholesterol levels, liver damage, kidney and testicular cancer etc.
- Difficult to capture and destroy.

Measures to reduce PFAS contamination:

- Safe disposal of PFAS using methods like filtration through activated carbon tanks, incineration etc.;
- phasing out PFASs listed under the Stockholm Convention and replace them with safer alternatives;
- Avoid PFAS-based non-stick pans and kitchen utensils; etc.

Nonylphenol (NP)

- Nonylphenol is commonly used in the production of **Nonylphenol Ethoxylates (NPEs)**.
- NPEs used as **surfactants** and in day-to-day consumer products such as detergents, wetting agents and dispersants.
- **A study revealed the presence of high levels of toxic chemical NP in drinking water across India.**
- **Impacts in Human:** Endocrine disruptor associated with adverse effects like birth defects, skin and eye irritation etc.
- **United Nations Environment Programme (UNEP)** has designated NP as a **chemical of global concern**. **No standards exclusively for NP** in drinking and surface waters **in India**.

Bioremediation

- Bioremediation uses microorganisms (bacteria and fungi) to degrade environmental contaminants into less toxic forms.
- Microorganisms can be specifically designed for bioremediation using genetic engineering techniques. **Can you recall?**
- The process of bioremediation can be monitored indirectly by measuring the **Oxidation Reduction Potential or redox in soil and groundwater, together with pH, temperature, oxygen content, electron acceptor/donor concentrations, and concentration of breakdown products** (e.g. carbon dioxide).

Kashmir's Dal Lake harbours bacteria that can degrade pesticides

The presence of naturally-occurring bacteria that is resistant to an insecticide widely used in the region may help in bioremediation of degrading water bodies

- Researchers have reported the presence of naturally occurring chloropyrifos- resistant bacteria (**E.Coli and Pseudomonas fluorescens**) in the environments of Kashmir Valley.
 - Chloropyrifos is an **organophosphorous insecticide** widely used in the region.
- The finding, according to the researchers, could lead to the use of microbial systems for **removal of pollutants from contaminated**.

In-situ (at the site) bioremediation

- **Bioventing:** Supply of nutrients through **wells to contaminated soil** to stimulate the growth of bacteria. It is used for **simple hydrocarbons** and can be used where the contamination is deep under the surface.
- **Biosparging:** Injection of air under pressure below the water table to increase groundwater oxygen concentrations and enhance the rate of biological degradation of contaminants by bacteria.
- **Bioaugmentation:** Microorganisms are imported to a contaminated site to enhance the degradation process.

Ex-situ bioremediation

- **Landfarming:** contaminated soil is excavated and spread over a prepared bed and periodically tilled until pollutants are degraded. The goal is to stimulate indigenous biodegradative microorganisms and facilitate their aerobic degradation of contaminants.
- **Bioreactors:** involve the processing of contaminated solid material (soil, sediment, sludge) or water through an engineered containment system.
- **Biopiles:** It is a hybrid of landfarming and composting. Essentially, engineered cells are constructed as aerated compost piles. Typically used for treatment of surface contamination with petroleum hydrocarbons.
- **Composting:** Composting is nature's recycling of decomposed organic materials into a rich soil known as compost.

Advantages of bioremediation

- Destroys a wide variety of contaminants.
- The destruction of target pollutants is possible.
- Less expensive and environment-friendly.

Disadvantages of bioremediation

- Bioremediation is limited to biodegradable compounds.
- Not all compounds are susceptible to biodegradation.
- It often takes a longer than other treatment processes

Phytoremediation

- Phytoremediation is the **use of plants to remove contaminants** from soil and water. Mangroves, estuarine vegetation and other wetland vegetation carry out natural phytoremediation.

Types:

- **Phytoextraction/phytoaccumulation:** It is the **accumulation of contaminants into the roots and aboveground shoots or leaves of plants**. E.g. **Water hyacinth (an aquatic weed, invasive species)** can purify water by taking some toxic materials and several heavy metals from water. **Planting eucalyptus trees** all along sewage ponds is suggested. These trees absorb all surplus wastewater rapidly and release pure water vapour into the atmosphere.

- **Phytotransformation or phytodegradation:** It refers to the uptake of organic contaminants from soil, sediments, or water and their transformation to more stable, less toxic, less mobile form.
- **Phytostabilization:** It is a technique in which plants reduce the mobility and migration of contaminated soil. Leachable constituents are adsorbed and bound into the plant structure so that they form unstable mass of plant from which the contaminants will not re-enter the environment.

- **Mycoremediation:** It is a form of bioremediation in which **fungi** are used to decontaminate the area.
- **Mycofiltration:** It is a similar process, using **fungal mycelia** to filter toxic waste and microorganisms from water in soil.

Stockholm Convention on POPs (UNEP convention)

- Stockholm Convention on Persistent Organic Pollutants (POPs) is an international treaty enacted in 2004 to eliminate or restrict the production and use of POPs.
- **Membership:** 186 parties (185 states and the European Union).
 - India ratified in 2006.
 - The Ministry of Environment notified the 'Regulation of POP Rules' in 2018, under the Environment (Protection) Act, 1986.
 - The chemicals targeted are listed in the annexes of the convention text.
- **Notable non-ratifying states:** the **United States**, (no surprises there) Israel, and Malaysia.

Persistent Organic Pollutants (POPs)

- POPs are defined as “chemical substances that **persist** in the environment, **bioaccumulate** through the food web, & pose a risk of causing adverse effects to human health & the environment”.
- The most commonly encountered POPs are **organochlorine pesticides**.

Why POPs are so harmful?

- They remain intact for exceptionally long periods of time (many years).
- They become widely distributed throughout the environment as a result of natural processes involving soil, water and, most notably, air, long range environment transport (LRET).

- They accumulate in the **fatty tissue** of living organisms including humans, and are found at higher concentrations at higher levels in the food chain;
- They are **toxic** to both humans and wildlife.
- Exposure to POPs can lead to cancer, damage to central & peripheral nervous systems, diseases of Endos immune system, reproductive disorders and interference with normal infant and child development.
- In addition, POPs concentrate in living organisms through another process called **bioaccumulation**. (upto 70000 times)

Included Listed PoPs	Use
Aldrin	insecticide.
Endrin	agricultural insecticide
Heptachlor	termiticide
Hexachlorobenzene	chemical intermediate and a solvent for pesticides.
DDT	chlorine-containing pesticides



Annexes of Stockholm Convention

 Categorization of chemicals	 Measures to be taken by parties with respect to listed chemicals
 Annex A (Elimination)	• Eliminate the production and use. • Specific exemptions available only to Parties registered for them.
 Annex B (Restriction)	• Restrict the production and use in light of any applicable acceptable purposes and/or specific exemptions.
 Annex C (Unintentional production)	• Reduce the unintentional releases with the goal of continuing minimization and, where feasible, ultimate elimination.



Related News:

- 18th Meeting of the Persistent Organic Pollutants Review Committee to the Stockholm Convention (POPRC-18) was held recently.
- **Outcomes of the meeting:**
 - Committee concluded its review of four of the five chemicals under consideration.
 - **The 5 chemicals:** Dechlorane Plus (flame retardant), UV-328 (stabiliser), medium chain chlorinated paraffins (flame retardant), Long-Chain Perfluorocarboxylic Acids, and chlorpyrifos (Pesticide).

Endosulfan

- An **organochlorine biocide** used for controlling **pests and mites** by **generating neurotoxic effects**. It is sprayed on **cotton, cashew, fruits, tea, paddy, and tobacco** etc.
- **Banned in 2011** by SC and under **Stockholm Convention on Persistent Organic Pollutants**.
- Complete ban w.e.f of December 2020 was also put on **Alachlor, Dichlorvos, Trichlorfon, Phosphamidon, methyl parathion, phorate and triazophos**.
- **Supreme Court (SC)** directed **Kerala** to pay ₹5 lakh compensation each victim of toxic Endosulfan pesticide. : Kasargod Tragedy

Nanoplastics

- **Context:** Researchers have developed a new, metallic fingerprint-based method to detect and measure amount of nanoplastics in organisms.
- Researchers exposed lettuce plants to nanoplastics from commonly found plastic waste in environment — polystyrene (PS) and polyvinyl chloride (PVC) nanoplastics.
- Nanoplastics are **smaller than 1,000 nanometre** (1 nm is equal to one billionth of a metre).
- It can lead to **cell damage and inflammation in humans**; growth impairments, larval deformities and other toxic effects in marine organisms.

Basel Convention on the Control of Transboundary Movement of hazardous wastes (UNEP)

- **The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal** was adopted on **22 March 1989** by the **Conference of Plenipotentiaries in Basel, Switzerland**, in response to a public outcry following the discovery, in the 1980s, in Africa and other parts of the developing world of deposits of toxic wastes imported from abroad.
- It entered into force in **1992**.
- As of **2018, 190 members** are parties to the Convention. The United States has signed the Convention but has not ratified it.
- India is a **member** of the Basel Convention. It ratified the convention in June 1992 and brought it into force on 22nd September

Basel Convention was designed to:

- Reduce the movement of hazardous waste between nations.
- Prevent the transfer of hazardous waste from developed to less developed countries (LDCs).
- Minimize the amount and toxicity of wastes generated.
- Assist LDCs in environmentally sound management of the hazardous and other wastes they generate.

- **Annexes:**
 - **Annex I of the Convention:** as further clarified in Annexes VIII and IX, lists those wastes that are classified as hazardous and subject to the control procedures under the Convention.
 - **Annex II of the Convention** identifies those wastes that require special consideration (known as "other wastes", and which primarily refer to household wastes).
- **Examples of Waste Included:** Biomedical and healthcare wastes, Used oils, Used lead acid batteries, Polychlorinated Biphenyls (PCBs).

Rotterdam Convention (UNEP)

- It was adopted in **1998** by a **Conference of Plenipotentiaries** in Rotterdam, the Netherlands and entered into force on 24 February 2004.
- The Convention creates legally binding obligations for the implementation of the **Prior Informed Consent (PIC) procedure**.
- It built on the **voluntary PIC procedure**, initiated by UNEP and FAO in 1989 and ceased on 24 February 2006.

- The Convention covers **pesticides and industrial chemicals** that have been banned or severely restricted for health or environmental reasons by Parties and which have been notified by Parties for inclusion in the PIC procedure.

Prior Informed Consent

- A mechanism under Rotterdam Convention for formally obtaining and disseminating the decisions of importing parties on their willingness to receive future shipments of hazardous chemicals listed in Annex III of the Convention and for ensuring compliance with these decisions by exporting Parties.

Objectives:

- to promote shared responsibility and cooperative efforts among Parties in the international trade of certain hazardous chemicals in order to protect human health and the environment from potential harm;
- to contribute to the environmentally sound use of those hazardous chemicals, by facilitating information exchange about their characteristics, by providing for a national decision-making process on their import and export and by disseminating these decisions to Parties.

Other key information:

- Covers pesticides and industrial chemicals that have been banned or severely restricted for health or environmental reasons by Parties.
- Creates **legally binding obligations** for the implementation of the **PIC procedure. Information Exchange is facilitated among Parties for a very broad range of potentially hazardous chemicals.**
- **A subsidiary body Chemical Review Committee (CRC)** was established to review chemicals and pesticide formulations according to criteria set out by Convention.

Annex III Chemicals

- The chemicals listed in Annex III include pesticides and industrial chemicals that have been banned or severely restricted for health or environmental reasons by two or more Parties and which the Conference of the Parties has decided to subject to the PIC procedure.
- There are a total of **43 chemicals listed in Annex III**, 32 are pesticides (including 4 severely hazardous pesticide formulations) and 11 industrial chemicals.

Related News

- Recommendations were made by the **Chemical Review Committee (CRC)** for 2 hazardous pesticides:
 - **Iprodione:** A fungicide used on vines, fruits, trees and vegetables, classified as carcinogenic and toxic for reproduction.
 - **Terbufos:** A soil insecticide used commonly on sorghum, maize, beet and potatoes, found to pose risk to aquatic organisms due to its toxicity.
- In India, the use of these chemicals was permitted by the **2015 Anupam Verma committee report**. India is among the largest exporters of **Terbufos**.

Other Related News

- According to a report by the **Pesticide Action Network (PAN)** a non-profit body, chlorpyrifos, fipronil, atrazine and paraquat are being used in excess in various states than what has been recommended by the **Central Insecticides Board and Registration Committee**, India's pesticides regulator.

Chlorpyrifos	to control foliage and soil-borne insect pests.
Fipronil	to control ants, beetles, cockroaches, fleas, etc.
Atrazine	to selectively control annual grasses and broadleaf weeds before they emerge.
Paraquat	Weedicide and herbicide

Hong Kong Convention on Recycling of Ships

- **The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009**, was developed in cooperation with **the International Labour Organization and the Parties to the Basel Convention**. The Hong Kong Convention intends to address all the issues around ship recycling, including the fact that ships sold for scrapping may contain hazardous substances such as asbestos, heavy metals etc.
- It also addresses concerns about the working conditions at many of the world's ship recycling locations.

India and Ship Recycling

- India is the **leader in the global ship recycling** industry, with a share of over 30% of the market. Around 800 vessels are sent for breaking every year around the world. India accounts for around 30% of that work, with an average 250 ships recycled majorly at centre in Alang, Gujarat.
- Under **Ship Recycling Act, 2019**, India has acceded to Hong Kong Convention for Ship Recycling under International Maritime Organization (IMO).

International Convention for the Prevention of Pollution from Ships (MARPOL)

- The International Convention for the Prevention of Pollution from Ships (MARPOL) is the **main international convention covering prevention of pollution** of the marine environment by ships from operational or accidental causes.
- The MARPOL Convention was adopted on 2 November 1973 at **IMO**.
- The Convention includes regulations aimed at preventing and minimizing pollution from ships - both accidental pollution and that from routine operations - and currently includes **six technical Annexes**.

IMO's International Code for Ships Operating in Polar Waters (Polar Code)

- The Polar Code covers the **full range of design, construction, equipment, operational, training, search and rescue and environmental protection matters** relevant to ships operating in the inhospitable waters surrounding the two poles. The Polar Code entered into force on 1 January **2017**.
- The Polar Code includes **mandatory measures** covering safety part (part I-A) and pollution prevention (part II-A) and recommendatory provisions for both (parts I-B and II-B).

- The Code requires ships intending to operating in the defined waters in Poles to apply for a **Polar Ship Certificate**, which classify the vessel as:
 - **Category A ship** - ships designed for operation in polar waters at least in medium first-year ice, which may include old ice inclusions;
 - **Category B ship** - a ship not included in category A, designed for operation in polar waters in at least thin first-year ice, which may include old ice inclusions; or
 - **Category C ship** - a ship designed to operate in open water or in ice conditions less severe than those included in Categories A and B.
- The issuance of a certificate would require an assessment, taking into account the anticipated range of operating conditions and hazards the ship may encounter in the polar waters.

London Convention (Convention on Dumping of Wastes at Sea)

- An inter-governmental conference on the Convention on the Dumping of Wastes at Sea met in London in 1972 to adopt this instrument, the **London Convention**. The Convention has a global character and is aimed at **international control and putting an end to marine pollution**.
- The definition of dumping under the Convention relates to the **deliberate disposal at sea of wastes or other materials from vessels, aircraft, platforms & other man-made structures**.

- 'Dumping' here does not cover wastes derived from the exploration & exploitation of sea-bed minerals.
- The 1978 amendment to the convention **banned the incineration of wastes** at sea.
- It **phased out the dumping of industrial wastes by 1995.**

Under2 Coalition

- The **Under2 Coalition** is a coalition of **subnational governments** that **aims** to achieve **greenhouse gases emissions mitigation**.
- The Under2 MOU was conceived through a partnership between the **governments of California and Baden-Wurttemberg**. The memorandum was developed just before the **Paris Agreement**.
- It was signed by **twelve founding jurisdictions** on **May 19, 2015** in **Sacramento, California**.
- Originally called the **Under2 MOU**, it became known as the Under2 Coalition in **2017**.
- As of **October 2022**, the list of signatories had grown to **270 governments**.

- The **Climate Group** acting as secretariat.
- **Purpose:** The intent of the memorandum signatories is for each to achieve Greenhouse gas emission reductions consistent with a trajectory of 80 to 95 percent below 1990 levels by 2050 and/or achieving a per capita annual emission goal of less than 2 metric tons by 2050.
- The Under2 MoU allows subnational governments such as cities, counties and states to highlight their work to reduce greenhouse gas emissions.

Blue Flag Certification



- This Certification is accorded by an international agency “Foundation for Environment Education, Denmark” based on 33 stringent criteria in four major heads i.e
 - Environmental Education and Information,
 - Bathing Water Quality,
 - Environment Management and Conservation and
 - Safety and Services on the beaches.
- It started in France in 1985 and has been implemented in Europe since 1987, and in areas outside Europe since 2001 when South Africa joined.
- Spain tops the list with 566 such beaches; Greece and France follow with 515 and 395, respectively.

Blue Flag Certification

BLUE FLAG BEACHES OF INDIA



Mangroves

- Mangroves are **salt-tolerant (halophytes)** evergreen shrubs or small trees (vary in height from 8 to 20 m) that represent a characteristic littoral (near the seashore) forest ecosystem.
- They grow below the highwater level of spring tides.
- The best locations are where **abundant silt** is brought down by rivers or on the backshore of accreting sandy beaches.
- Such locations include **brackish waters** of sheltered low-lying coasts, estuaries, mudflats, tidal creeks, backwaters (coastal waters held back on land), marshes and lagoons of tropical and subtropical regions.



Adaptive Mechanism of Mangroves

- **Succulent leaves** (thick leaves adapted to store water and reduce evapotranspiration),
- **Sunken stomata** (to protect from drying winds),
- **Leaves with salt-secreting glands** (to flush out the excess salt)
- **Aerial breathing roots** called pneumatophores,
- **Vivipary** (seeds or embryos begin to develop before they detach from the parent),
- **Stilt and prop roots** (they are fibrous (adventitious) support roots)
- **Buttresses** (large, wide support roots on all sides of a shallowly rooted (mangrove) tree).



Adaptive Mechanism of Mangroves



- The **adventitious roots** (prop and stilt roots), which emerge from the main trunk above ground level, act as support roots. The complex root system helps mangroves overcome the strong wave action and diurnal tidal inundation.
- The **pneumatophores** (blind roots), prop roots and stilt roots help mangroves overcome the respiration problem in anaerobic (low oxygen — anoxic) soil conditions (a classic example of adaptation).
- Mangroves exhibit a **viviparity mode of reproduction**, i.e., seeds germinate in the tree itself (before falling to the ground). This is an adaptive mechanism to overcome the problem of germination in saline water.

Why are mangroves not found beyond subtropics?



- Mangroves are confined to the tropics and subtropics, occurring mainly in the intertidal regions between latitudes 24° N and 38° S.
- This is because the mangrove vegetation contains a **complex salt filtration system that facilitates high water loss at the root level itself to cope with the brackish conditions.**
- These adaptive mechanisms are energy-intensive and require high solar radiation.

Mangrove Cover in India



Table 3.2 Mangrove Cover Assessment 2021

(in sq km)

Sl. No.	State/UT	Very Dense Mangrove	Moderately Dense Mangrove	Open Mangrove	Total	Change with respect to ISFR 2019
1.	Andhra Pradesh	0	213	192	405	1
2.	Goa	0	21	6	27	1
3.	Gujarat	0	169	1,006	1,175	-2
4.	Karnataka	0	2	11	13	3
5.	Kerala	0	5	4	9	0
6.	Maharashtra	0	90	234	324	4
7.	Odisha	81	94	84	259	8
8.	Tamil Nadu	1	27	17	45	0
9.	West Bengal	994	692	428	2,114	2
10.	AGN Islands	399	168	49	616	0
11.	D&NH and Daman & Diu	0	0	3	3	0
12.	Puducherry	0	0	2	2	0
Total		1,475	1,481	2,036	4,992	17

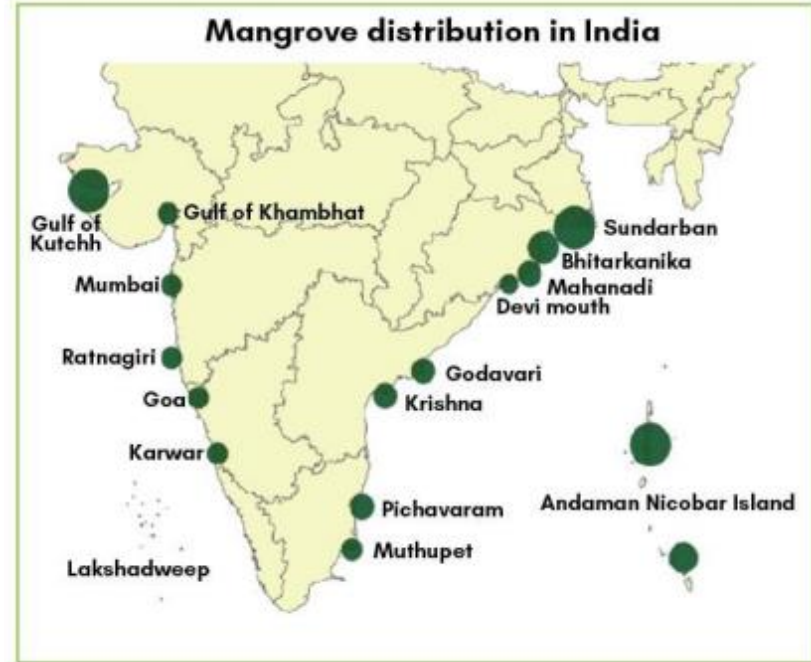
Mangrove Cover in India

- The mangroves of Sundarbans are the largest single block of tidal halophytic mangroves of the world. (Sundari tree)
This mangrove forest is famous for the Royal Bengal Tiger and crocodiles.
- **Bhitarkanika (Orissa):** Second largest in the Indian sub-continent
- Mangrove swamps also occur in profusion in the intertidal mudflats on both side of the creeks in the Godavari-Krishna deltaic regions of Andhra Pradesh.



Mangrove Cover in India

- On the west coast of India, mangroves, mostly scrubby and degraded occur along the intertidal region of estuaries and creeks in Maharashtra, Goa and Karnataka.
- The mangrove vegetation in the coastal zone of Kerala is very sparse and thin.
- In Gujarat (north-west coast) mangroves *Avicennia marine*, *Avicennia officinalis* and *Rhizophora mucronata* are found mainly in Gulf of Kutch and the Kori creek.

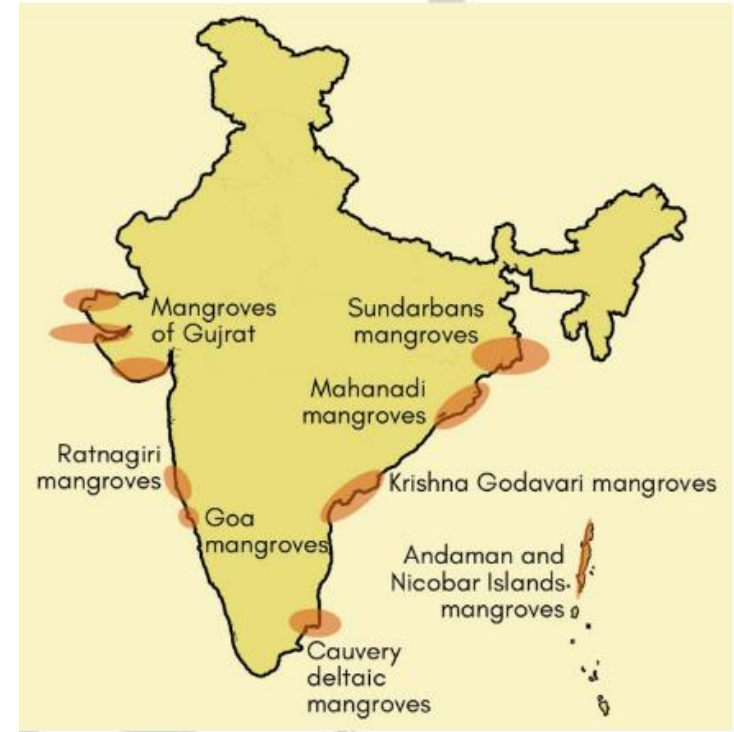


State of Mangroves Report 2022

Released by: Global Mangrove Alliance
(annual report).

Key findings

- Global Mangrove extent is of 147,359 km² (2020). It is higher than previous estimates of 136,000 km² (for 2016) largely due to improved maps rather than real gains.
- The rates of loss have greatly diminished, with averaged losses over the last decade of just 66 km² of all mangroves per year (327 km² between 1996 and 2010).



State of Mangroves Report 2022



GLOBAL
MANGROVE
ALLIANCE

Global Mangrove Alliance (GMA)



Genesis: Launched in 2018 at the World Ocean Summit by Conservation International (CI), the International Union for Conservation of Nature (IUCN), The Nature Conservancy (TNC), Wetlands International, and World Wildlife Fund (WWF).



Objective: To increase the global area of mangrove habitat through conservation, restoration and equitable management.

➤ This includes **supporting research, advocacy, education and practical projects** on the ground with **local and community partners** with focus on 3 key areas:

- Halting Mangrove Loss;
- Implementing science-based restoration; and
- Building Awareness.



Membership: Includes **more than 30 organizations**, coordinated by members Conservation International, The International Union for the Conservation of Nature, The Nature Conservancy, Wetlands International and World Wildlife Fund.



Other key information:

- Published the inaugural 'The State of the World's Mangroves 2021'.
- Runs a number of initiatives for Mangroves such as 'Mangroves for the Future' (MFF) - a unique partner-led initiative to promote investment in coastal ecosystem conservation for sustainable development.
 - MFF covers Bangladesh, Cambodia, India, Indonesia, Maldives, Myanmar, Pakistan, Seychelles, Sri Lanka, Thailand and Vietnam.

Initiatives for Mangrove



- **Mangrove Alliance for Climate (MAC)** (Launched by The United Arab Emirates (UAE) and Indonesia.
- Other members include Sri Lanka, Australia, Japan, **India** and Spain.
- Launched during COP27

- **Building with Nature in Indonesia (launched by Indonesia)**
- Aims to naturally regenerate mangroves and protect Indonesia's coast against flooding.
- Recognised under **World Restoration Flagships** (UN Decade on Ecosystem Restoration declared to the best examples of large-scale and
- long-term ecosystem restoration in world)

- **MISHTI scheme (Mangrove Initiative for Shoreline Habitats & Tangible Incomesa):** Launched under Budget 2023-24
- Take up mangrove plantation along the coastline and on salt pan lands, wherever feasible, through convergence between **MGNREGS**, **CAMPA Fund** and other sources.
- **Sustainable Aquaculture in Mangrove Ecosystem (SAIME) Initiative**
- It is new initiative of sustainable shrimp cultivation provides hope for mangrove restoration in Sundarbans.
- It is a community-based pilot project in West Bengal under which farmers are planting mangrove trees around shrimp ponds.
- Generally, mangrove forests are cleared to cultivate shrimps in the areas. o It is being conceived by Nature Environment and Wildlife Society (NEWS), Global Nature Fund and others.

Importance of Mangrove



- Highly productive ecosystems with rich biodiversity.
- Very efficient in dissipating the sea wave energy, thus protecting the coastal areas from tsunamis, **storm surges (produced by cyclones)** and soil erosion.
- Slow down water flow and act as a zone of land accretion by enhancing sediment deposition.
- Act as a **riparian buffer** and trap pollutants, including heavy metal contaminants. They enhance the natural recycling of nutrients.
- Essential **carbon sink**.
- Act as an essential source of **livelihood** for the coastal communities dependent on the collection of honey, tannins, wax, firewood, medicinal plants, edible plants, and fishing

Threats to Mangroves



Natural threats:

- Cyclones, typhoons and strong wave actions.
- Trampling and overgrazing by wildlife and livestock close to mangrove regions.
- Damage by crabs, oysters and pests to the young seedlings of mangroves.

Anthropological threats:

- **Agriculture:** Many thousands of acres of mangrove forest have been destroyed to make way for rice paddies, rubber trees, palm oil plantations, and other forms of agriculture.
- **Coastal Development**
- **Shrimp Farming:** By far the greatest threat to the world's mangrove forests is the rapidly expanding shrimp aquaculture industry.

2013

Which of the following can be found as pollutants in the drinking water in some parts of India?

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below.

- a) 1 and 3 only
- b) 2, 4 and 5 only
- c) 1, 3 and 5 only
- d) 1, 2, 3, 4 and 5

Answer: C

- Water pollutants include contamination due to domestic wastes, insecticides and herbicides, food processing waste, pollutants from livestock operations, volatile organic compounds (VOCs), heavy metals, chemical waste, and others.
- **Uranium** in groundwater is indeed a **problem in Punjab** and **leading to the birth of mentally retarded children.**
- **Arsenic and Fluoride** are well-known **water pollutants.**

2019

In the context of which one of the following are the terms ‘pyrolysis and plasma gasification’ mentioned?

- a) Extraction of rare earth elements
- b) Natural gas extraction technologies
- c) Hydrogen fuel-based automobiles
- d) Waste-to-energy technologies

Answer: D

Explanation:

- Pyrolysis is the process of heating organic material at high temperatures in the absence of oxygen.
- Plasma gasification is an extreme thermal process using plasma that converts organic matter into syngas (synthesis gas) which is primarily made up of hydrogen and carbon monoxide. **Hence, option d is correct.**

2013

Acid rain is caused by the pollution of the environment by -

- a) Carbon dioxide and nitrogen
- b) Carbon monoxide and carbon dioxide
- c) Ozone and carbon dioxide
- d) Nitrous oxide and sulphur dioxide

Answer: D

- When **oxides of nitrogen and sulphur** mix with water in the atmosphere and come down in the form of rain, it is termed as acid rain.
- Acid rain is caused by a chemical reaction that begins when compounds like sulphur dioxide and nitrogen oxides are released into the air. These substances can rise very high into the atmosphere, where they mix and react with water, oxygen, and other chemicals to form more acidic pollutants, known as acid rain.

2013

Photochemical smog is a resultant of the reaction among -

- a) NO_2 , O_3 and peroxyacetyl nitrate in the presence of sunlight
- b) CO , O_2 and peroxyacetyl nitrate in the presence of sunlight
- c) CO , CO_2 , and NO_2 at low-temperature
- d) High concentration of NO_2 , O_3 , and CO in the evening

Answer: A

- Smog is a secondary pollutant, which is made up of Smoke and Fog. It is caused by a mixture of smoke and sulfur dioxide. The major chemical pollutants in Photochemical smog are NO, NO₂, volatile organic compounds, ozone, and PAN Perooxyacetyl Nitrate. **Hence, option a is correct.**

2013

Due to improper/indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste?

1. Beryllium
2. Cadmium
3. Chromium
4. Heptachlor
5. Mercury
6. Lead
7. Plutonium

Select the correct answer using the codes given below.

- a) 1, 3, 4, 6 and 7 only
- b) 1, 2, 3, 5 and 6 only
- c) 2, 4, 5 and 7 only
- d) 1, 2, 3, 4, 5, 6 and 7

Answer: B

- The disposed of electrical or electronic devices are termed electronic waste or e-waste. The elements of e-waste are lead, mercury, cadmium, chromium, beryllium, Polyvinyl chloride, Sulphur, Perfluorooctanoic, and Brominated Flame Retardant.

2013

Which one among the following industries is the maximum consumer of water in India?

- a) Engineering
- b) Paper and pulp
- c) Textiles
- d) Thermal power

Answer: D

- Of the total water use by the industry, **thermal power plants are the biggest users of water and account for 88%** of the total industrial water use. They are followed by engineering (5.05%), pulp & paper (2.26%), and textiles (2.07%) industries.

2014

Which of the following are some important pollutants released by the steel industry in India?

1. Oxides of sulphur
2. Oxides of nitrogen
3. Carbon monoxide
4. Carbon dioxide

Select the correct answer using the code given below.

- a) 1, 3 and 4 only
- b) 2 and 3 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Answer: D

- In a steel furnace, coke reacts with iron ore, releasing iron and generating CO and CO₂ gases. Due to the use of coal, pollutants such as SO_x and NO_x are released, therefore all are correct options.

2016

In the cities of our country, which among the following atmospheric gases are normally considered in calculating the value of Air Quality Index?

1. Carbon dioxide
2. Carbon monoxide
3. Nitrogen dioxide
4. Sulphur Dioxide
5. Methane

Select the correct answer using the code given below

- a) 1, 2 and 3 only
- b) 2, 3 and 4 Only
- c) 1, 4 and 5 Only
- d) 1, 2, 3, 4 and 5

Answer: B

The AQI is an index for reporting daily air quality.

- It tells you how clean or polluted your air is, and what associated health effects might be a concern for you.
- The AQI focuses on health effects you may experience within a few hours or days after breathing polluted air.
- EPA calculates the AQI for five major air pollutants regulated by the Clean Air Act: **ground-level ozone, particle pollution (also known as particulate matter), carbon monoxide, sulfur dioxide, and nitrogen dioxide.**
- For each of these pollutants, EPA has established national air quality standards to protect public health.
- Ground-level ozone and airborne particles are the two pollutants that pose the greatest threat to human health in this country.

2017

Biological Oxygen Demand (BOD) is a standard criterion for -

- a) Measuring oxygen levels in the blood
- b) Computing oxygen levels in forest ecosystems
- c) Pollution assay in aquatic ecosystems
- d) Assessing oxygen levels in high altitude regions

Answer: C

- Biological oxygen demand (BOD) is the amount of dissolved oxygen needed (i. e., demanded) by aerobic microorganisms to break down organic material present in a given water sample. Highly polluted water will have a high level of biological oxygen demand (BOD). **Hence, option c is correct.**
- It is the amount of dissolved oxygen needed by bacteria in decomposing the organic wastes present in water. It is expressed in milligrams of oxygen per liter of water.
- The higher value of BOD indicates low DO content of water.

2017

In the context of solving pollution problems, what is/are the advantage/advantages of the bioremediation technique?

1. It is a technique for cleaning up pollution by enhancing the same biodegradation process that occurs in nature.
2. Any contaminant with heavy metals such as cadmium and lead can be readily and completely treated by bioremediation using microorganisms.
3. Genetic engineering can be used to create microorganisms specifically designed for bioremediation.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: C

- Bioremediation is a treatment that uses naturally occurring organisms to break down hazardous substances into less toxic or non-toxic substances. It uses **microorganisms** to degrade organic contaminants in soil, groundwater, sludge, and solids. The microorganisms break down contaminants by using them as an energy source or metabolizing them with an energy source. **Hence, statement 1 is correct.**
- Not all contaminants are easily treated by bioremediation using microorganisms. For example, heavy metals such as cadmium and lead are not readily absorbed or captured by microorganisms. **Hence, statement 2 is incorrect.**

- Genetic engineering has been used to create organisms designed for specific purposes. For e.g. bacterium *Deinococcus radiodurans* (the most radioresistant organism known) has been modified to consume and digest toluene and ionic mercury from highly radioactive nuclear waste. **Hence, statement 3 is correct.**

2018

In India, 'extended producer responsibility' was introduced as an important feature in which of the following?

- a) The Bio-medical Waste (Management and Handling) Rules, 1998
- b) The Recycled Plastic (Manufacturing and Usage) Rules, 1999
- c) The e-Waste (Management and Handling) Rules, 2011
- d) The Food Safety and Standard Regulations, 2011

Answer: C

E-Waste (Management and Handling) Rules, 2011

- Extended producer's responsibility (EPR) is the main feature of the E-waste (Management and Handling) Rules, 2011, wherein the producer of electrical and electronic equipment has the responsibility of managing such equipment after its 'end of life', thus the producer is responsible for their products once the consumer discards them. **Hence, option c is correct.**

2019

As per the Solid Waste Management Rules, 2016 in India, which one of the following statements is correct?

- a) Waste generator has to segregate waste into five categories.
- b) The Rules are applicable to notified urban local bodies, notified towns, and all industrial townships only.
- c) The Rules provide exact and elaborate criteria for the identification of sites for landfills and waste processing facilities.
- d) It is mandatory on the part of the waste generator that the waste generated in one district cannot be moved to another district.

Answer: C

- The SWM Rules 2016 provide detailed criteria for setting -up solid waste processing and treatment facility, solid waste management in hilly areas, for waste to energy process, Sanitary Landfills, site selection, development of facilities at the sanitary landfills, specifications for landfilling operations and closure on completion of landfilling, pollution prevention, Closure and Rehabilitation of Old Dumps, etc. **Hence, option c is correct.**

2019

Why is there a great concern about the ‘microbeads’ that are released into the environment?

- a) They are considered harmful to marine ecosystems.
- b) They are considered to cause skin cancer in children.
- c) They are small enough to be absorbed by crop plants in irrigated fields.
- d) They are often found to be used as food adulterants.

Answer: A

- Microbeads are manufactured solid plastic particles of less than one millimeter in their largest dimension
- They are most frequently made of polyethylene but can be of other petrochemical plastics such as polypropylene and polystyrene.
- Microbeads, small pellets of plastic, extensively used in personal care products such as shampoo, baby lotion, and face cream and considered toxic to marine life, are being banned internationally. **Hence, option a is correct.**

2019

Consider the following:

1. Carbon monoxide
2. Methane
3. Ozone
4. Sulphur dioxide

Which of the above are released into the atmosphere due to the burning of crop/biomass residue?

- a) 1 and 2 only
- b) 2, 3 and 4 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Answer: D

- Agricultural crop residue burning contributes towards the emission of greenhouse gases (CO_2 , N_2O , CH_4), air pollutants
- (CO , NH_3 , NO_x , SO_2 , NMHC, volatile organic compounds), particulate matter and smoke thereby posing a threat to human health. **Hence, option d is correct.**

2020

Which of the following are the reasons/factors for exposure to benzene pollution?

1. Automobile exhaust
2. Tobacco smoke
3. Wood burning
4. Using varnished wooden furniture
5. Using products made of polyurethane

Select the correct answer using the code is given below:

- a) 1, 2 and 3 only
- b) 2 and 4 only
- c) 1, 3 and 4 only
- d) 1, 2, 3, 4 and 5

Answer: A

- Sources of exposure to benzene Human exposure to benzene has been associated with a range of acute and long-term adverse health effects and diseases, including cancer and aplastic anemia. Active and passive exposure to tobacco smoke is also a significant source of exposure. Benzene is highly volatile, and most exposure is through inhalation. Natural sources of benzene include volcanoes and forest fires. Benzene is also a natural part of crude oil, gasoline, and cigarette smoke.
- Industrial processes As benzene occurs naturally in crude petroleum at levels up to 4 g/l, human activities using petroleum lead to exposure. These activities include processing petroleum products, coking of coal, production of toluene, xylene, and other aromatic compounds, and use in industrial and consumer products, as a chemical intermediate and as a component of petrol (gasoline) and heating oils.

- The presence of benzene in petrol and as a widely used industrial solvent can result in a significant occupational exposure and widespread emissions to the environment. Automobile exhaust accounts for the largest source of benzene in the general environment. Wood finishes can contain and emit various chemicals. For example, paints and some wood treatments can contain formaldehyde, acetone, toluene, or butanol. Wood dyes can contain nonane, decane, undecane, dimethyloctane, dimethylnonane, trimethylnonane, trimethylbenzene. Carbon monoxide is the most common, but smoke also contains benzene, toluene, nitrogen oxides, and hydrogen cyanide.

2020

Consider the following statements:

1. Coal ash contains arsenic, lead, and mercury.
2. Coal-fired power plants release Sulphur dioxide and oxides of nitrogen into the environment.
3. High ash content is observed in Indian coal.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Answer: D

- **Statement 1 is correct:** Coal itself isn't a particularly toxic material. But after it's burned, what remains in the ash includes lead, mercury, cadmium, chromium, arsenic, and selenium, all at levels that may threaten human health.
- **Statement 2 is correct:** Thermal power plants produce large amounts of nitrogen oxides and sulfur dioxide—the pollutants that cause acid rain—when they burn fossil fuels, especially coal, to produce energy.
- **Statement 3 is correct:** India's domestic coal reserves have a high ash content—up to 40 to 45 percent.

2021

Magnetite particles, suspected to cause neurodegenerative problems, are generated as environmental pollutants from which of the following?

1. Brakes of motor vehicles
2. Engines of motor vehicles
3. Microwave stoves within homes
4. Power plants
5. Telephone lines

Select the correct answer using the code given below.

- a) 1, 2, 3 and 5 only
- b) 1, 2 and 4 only
- c) 3, 4 and 5 only
- d) 1, 2, 3, 4 and 5 only

Answer: B

- Magnetite particles can act as pollutants and lead to neurodegenerative problems. Magnetic particles can penetrate and damage the blood-brain barrier and lead to the early development of **Alzheimer's disease**.
- Whenever any magnetic component is used in appliances, they generate magnetic flux which produces magnetic particles.
- **Option 1 is correct.** The automotive industry uses ceramic or ferrite magnets to make vehicles safer. One of the most impressive uses is in the Anti-lock Braking System (ABS) of vehicles. This use of magnets in motor vehicles can generate magnetic particles.
- **Option 2 is correct.** Engines of motor vehicles can generate magnetic particles as they use high-powered magnets which generate magnetic flux.

- **Option 4 is correct.** Power Plants release metal content along with ash. The use of coal in power plants is a major source of combustion-associated magnetite fine particles.

2021

Why is there a concern about copper smelting plants?

1. They may release lethal quantities of carbon monoxide into the environment.
2. The copper slag can cause the leaching of some heavy metals into the environment.
3. They may release sulphur dioxide as a pollutant.

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: B

- **Statement 1 is incorrect.** In the copper smelting process, first, the carbon (C) combusts with oxygen (O_2) in the air to produce carbon monoxide (CO). Second, the carbon monoxide reacts with the ore and removes one of its oxygen atoms, releasing carbon dioxide. Hence, it does not release a lethal quantity of CO into the environment.
- **Statement 2 is correct.** Copper smelters emit staggering amounts of toxic pollution i.e., lead, arsenic and selenium. These heavy metals and particulate matter from copper smelters contaminate the environment downwind and downstream.
- **Statement 3 is correct.** Copper smelters are the largest source of sulphur oxides (SO_x) and trace elements. Suspended particles have a pollution scope of 2-3 km, while Sulphur dioxide has a range of as much as 15 km.

2022

Consider the following

1. Carbon monoxide
2. Nitrogen oxide
3. Ozone
4. Sulphur dioxide

Excess of which of the above in the environment is/are the cause(s) of acid rain?

- a) 1, 2 and 3
- b) 2 and 4 only
- c) 4 only
- d) 1, 3 and 4

Answer: B (2 and 4 only)

- Photo oxidants like O_3 catalyse acid rain formation, they are not the primary causative agents.

2022

In the context of WHO Air Quality Guidelines, consider the following statements:

1. The 24-hr mean of PM_{2.5} should not exceed 15 $\mu\text{g}/\text{m}^3$ & annual mean of PM_{2.5} should not exceed 5 $\mu\text{g}/\text{m}^3$.
2. In a year, the highest levels of ozone pollution occur during the periods of inclement weather.
3. PM₁₀ can penetrate the lung barrier and enter the bloodstream.
4. Excessive ozone in the air can trigger asthma.

Which of the statements given above are correct?

- a) 1, 3 and 4
- b) 1 and 4 only
- c) 2, 3 and 4
- d) 1 and 2 only.

Answer: b) 1 and 4 only

- According to WHO Air Quality Guidelines, the annual average concentrations of PM_{2.5} should not exceed 5 $\mu\text{g}/\text{m}^3$, while 24-hour average exposures should not exceed 15 $\mu\text{g}/\text{m}^3$ more than 3 - 4 days per year. **So, statement 1 is correct.**
- Afternoon thunderstorm clouds block sunlight, causing ozone production to slow down for the day, while moisture from the storm destroys the ozone that has formed. **So, statement 2 is not correct.**
- Studies have shown that PM with a diameter of 10 μm (PM₁₀) can enter the lungs but not the bloodstreams. **So, statement 3 is not correct**
- Excessive ozone can trigger asthma because it is very irritating to the lungs and airways. **So, statement 4 is correct.**

2020

Steel slag can be the material for which of the following?

1. Construction of base road
2. Improvement of agricultural soil
3. Production of cement

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer- D)

- Steel slag, a by-product of steel making, is produced during the separation of the molten steel from impurities in steel-making furnaces. The slag occurs as a molten liquid melt and is a complex solution of silicates and oxides that solidifies upon cooling. It is used in various activities such as in road construction, these are the preferred aggregates as they harden well, enabling a thinner layer of pavement than what can be achieved with crushed stone aggregates. Slag is also very hardy, resistant to both weather and wear. **So, statement 1 is correct.**
- Benefits of using steel slag for crops: Slag application favors the increase of pH and the availability of nutrients such as Ca, Mg, and Si in the soil, which leads to the increase in the absorption of these elements by the plant, favoring the growth and yield of the crops. **So, statement 2 is correct.**
- Steel-slag can be used to produce energy-saving cement by co-grinding with OPC clinker and blast furnace slag. It can replace Portland cement in various applications and is especially suitable for projects where a low heat of hydration is required. **So, statement 3 is correct.**

2022

With reference to polyethylene terephthalate, the use of which is so widespread in our daily lives, consider the following statements:

1. Its fibres can be blended with wool and cotton fibres to reinforce their properties.
2. Containers made of it can be used to store any alcoholic beverage.
3. Bottles made of it can be recycled into other products.
4. Articles made of it can be easily disposed of by incineration without causing greenhouse gas emissions.

Which of the statements given above are correct?

- a) 1 and 3
- b) 2 and 4
- c) 1 and. 4
- d) 2 and 3

Answer: A (1 and 3 only)

- **Explanation:**

- Glass bottles are the safest when it comes to storing most liquids. Plastics have certain chemical substances (ethylene glycol and terephthalic acid), which tend to leach if alcohol is stored for a more extended period.

2022

Which one of the following statements best describes the ‘Polar Code’?

- a) It is the international code of safety for ships operating in polar waters.
- b) It is the agreement of the countries around the North Pole regarding the demarcation of their territories in the polar region.
- c) It is a set of norms to be followed by the countries whose scientists undertake research studies In the North Pole and South Pole.
- d) It is a trade and security agreement of the member countries of the Arctic Council.

Answer: A

- The International Code for Ships Operating in Polar Waters (the Polar Code) is a new code adopted by the IMO. The Code acknowledges that polar waters may impose additional demands on ships beyond those normally encountered. It provides a mandatory framework for ships operating in polar waters.
- It is mandatory under both the International Convention for the Safety of Life at Sea (SOLAS) and the International Convention for the Prevention of Pollution from Ships (MARPOL). The Polar Code covers the full range of design, construction, equipment, operational, training, search and rescue and environmental protection matters relevant to ships operating in the inhospitable waters surrounding the two poles. The Polar Code entered into force on 1 January 2017.

2022

Consider the following statements:

1. “The Climate Group” is an international non-profit organization that drives climate action by building large networks and runs them.
2. The International Energy Agency in partnership with the Climate Group launched a global initiative “EP100”.
3. EP100 brings together leading companies committed to driving innovation in energy efficiency and increasing competitiveness while delivering on emission reduction goals.
4. Some Indian companies are members of EP100.
5. The International Energy Agency is the Secretariat to the “Under2 Coalition”.

Which of the statements given above are correct?

- a) 1, 2, 4 and 5
- b) 1, 3 and 4 only
- c) 2, 3 and 5 only
- d) 1, 2, 3, 4 and 5

Answer: B

- **Statement 1 is correct:** Climate group is an international non-profit founded in 2003, with offices in London, New York, New Delhi, Amsterdam and Beijing. It has grown its network to include over 500 multinational businesses in 175 markets worldwide.
- **Statement 2 is incorrect:** EP100 is a global initiative led by The Climate Group and the Alliance to Save Energy.(and not by International Energy Agency)
- **Statement 3 is correct:** The Climate Group's global EP100 initiative brings together a growing group of energy-smart companies committed to improving their energy productivity and doing more with less greenhouse gas emission. By integrating ambitious energy targets into business strategy, leading companies are driving innovation in energy efficiency and increasing competitiveness while delivering on emissions reduction goals.

- **Statement 4 is correct:** EP100 has a global reach, with member companies operating in more than 130 markets around the world. India's own Mahindra group is one of the leaders in the ongoing battle to cut energy consumption and carbon dioxide emissions by increasingly switching to renewable energy.
- **Statement 5 is incorrect:** Climate Group is the Secretariat to the Under2 Coalition and not the International Energy Agency and works with governments to accelerate climate action through four work streams:
 - Pathways
 - Policy action
 - Transparency
 - Diplomacy

2021

With reference to furnace oil, consider the following statements:

- 1) It is a product of oil refineries.
- 2) Some industries use it to generate power.
- 3) Its use causes sulphur emissions into Environment.

Which of the statements given above are correct?

- a) 1 & 2 Only
- b) 2 & 3 Only
- c) 1 & 3 Only
- d) 1, 2 & 3

Answer: D) 1,2 & 3

Furnace oil or fuel oil is a dark viscous residual product of crude-oil distillation. It is used as a fuel in different types of combustion equipment. The emissions of oxides of sulphur are a direct result of the sulphur content of the fuel oil. **Hence, statements 1 and 3 are correct.**

Applications of Furnace oil:

- Marine engines and slow speed engines for power generation,
- Drying tea leaves,
- Gas turbines for power generation,
- Feedstock for fertiliser manufacturing,
- Thermic fluid heaters and hot air generators.

Hence, statement 2 is correct.

2020

In rural road construction, the use of which of the following is preferred for ensuring environmental sustainability or to reduce carbon footprint?

1. Copper slag
2. Cold mix asphalt technology
3. Geotextiles
4. Hot mix asphalt technology
5. Portland cement

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only
- (b) 2, 3 and 4 only
- (c) 4 and 5 only
- (d) 1 and 5 only

Answer: A) 1,2 & 3 only

The usage of Copper slag reduces the usage of primary materials as well as reduces the construction depth which in turn reduces energy demand in building. So, **(1) is correct.**

Cold asphalt mix is produced by mixing unheated mineral aggregate with either emulsified bitumen or foamed bitumen. Unlike hot mix asphalt (HMA), cold asphalt mix does not require any heating of aggregate which makes it economical and relatively pollution-free (no objectionable fumes or odours). So, **(2) is correct and (4) is not correct.**

Geotextiles reinforces the soil by adding tensile strength to it. It is used as a rapid de watering layer in the roadbed. So, **(3) is correct.**

Portland cement production needs energy and can impact the environment. So, **(5) is not correct.**

2011-12

The acidification of oceans is increasing. Why is this phenomenon a cause of concern?

- 1) The growth and survival of calcareous phytoplankton will be adversely affected.
- 2) The growth and survival of coral reefs will be adversely affected.
- 3) The survival of some animals that have phytoplanktonic larvae will be adversely affected.
- 4) The cloud seeding and formation of clouds will be adversely affected.

Which of the statement(s) given above is/are correct?

- a) 1, 2 and 3 only
- b) 2 only
- c) 1 and 3 only
- d) 1, 2, 3 and 4

Answer: D (1, 2, 3 and 4)

- **Explanation:** Ocean acidification decreases the calcifying ability of corals, calcareous plankton, crustaceans etc., and adversely affects the aquatic food webs. It also adversely affects cloud formation and cloud seeding.

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